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Prolonged Learning and Hasty Stopping: The Wald Problem with Ambiguity†

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This paper studies sequential information acquisition by an ambiguity-averse decision-maker (*DM*), *who decides how long to collect information before taking an irreversible action. The agent optimizes against the worst-case belief and updates prior by prior. We show that the consideration of ambiguity gives rise to rich dynamics: compared to the Bayesian DM*, *the DM here tends to experiment excessively when facing modest uncertainty and*, *to counteract it*, *may stop experimenting prematurely when facing high uncertainty. In the latter case*, *the DM's stopping rule is nonmonotonic in beliefs and features randomized stopping.* (*JEL* C61, D81, D83, D91)

The problem of making a decision on an action after deliberating on its merits is ubiquitous in many situations of life. Examples include a firm contemplating a potential new project, a jury deliberating on a verdict, a citizen pondering over his/her political vote, or a scientist evaluating a hypothesis. The pervasiveness of such a problem makes the framework of central importance, as seen by the extensive research on the subject matter; see Wald (1947); Arrow, Blackwell, and Girshick (1949); Moscarini and Smith (2001); Fudenberg, Strack, and Strzalecki (2018); Che and Mierendorff (2019), among others. With the exception of a few recent papers, the vast literature studying the problem considers a Bayesian framework. The Bayesian model misses, however, a crucial aspect of the problem: the decision-maker (DM) often lacks a clear model or beliefs about the uncertainty she is facing. In fact, Wald, often regarded as the first to formally introduce the framework, was mindful of the ambiguity and suggested a robust approach (Wald 1947), which anticipates the maxmin decision rule axiomatized by Gilboa and Schmeidler (1989).

We adopt the model in Gilboa and Schmeidler (1989) by considering a DM who has ambiguous beliefs about the state of the world and evaluates her choices against the worst-case scenario in a set of plausible priors. Specifically, the state is either

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L or R, and the DM must decide between two irreversible actions, ℓ and r. At each point, the DM can stop and choose an action. Alternatively, she can postpone her action and learn about the state. In the baseline model, the learning takes the form of a Poisson breakthrough that confirms state R. Imagine, for instance, a CEO of a biotech company seeking to establish the efficacy of a new drug or a theorist seeking to prove a theorem. The Bayesian optimal strategy in such a situation calls for looking for evidence of state R—efficacy of the drug or validity of the proof—for a duration of time until a long stretch of unsuccessful attempts convinces her to declare R to be unlikely and stop for action ℓ.

Introducing ambiguity in a dynamic setting raises conceptual issues about updating and dynamic consistency. To avoid the latter, the existing literature often follows a recursive approach, requiring the DM to adopt a prior-by-prior updating rule on a *rectangular* set of priors (see Epstein and Schneider 2003). Rectangularity requires that the set of priors is rich enough to contain a single worst-case belief that guides the DM's conditional choices. We follow a different approach: rather than assuming away dynamic inconsistency, we seek to understand its behavioral implications for a "sophisticated" DM who recognizes potential preference reversals and deals with them in a forward-looking and rational manner. To this end, we maintain prior-by-prior updating but allow for changes in the conditional worst-case scenario.

The paper shows that the consideration of ambiguity and time inconsistency gives rise to rich dynamics in sequential learning. Late in the process, ambiguity increases the DM's incentives to confirm the state, thereby prolonging experimentation. After a period of unsuccessful search for R-evidence, the DM contemplates two possibilities: no such evidence exists because the state is L or instead the state is R and she has simply been unlucky. As time progresses, the DM becomes increasingly convinced of the former. Following a sufficiently long period of futile search, the Bayesian DM would thus stop to take action ℓ. The ambiguity-averse DM, on the other hand, continues to worry about the second possibility and, as a consequence, keeps looking for more proof. Ambiguity aversion thus causes the DM to continue experimenting well past the time at which the Bayesian counterpart would stop. In the context of our example, the biotech CEO or the theorist refuses to see the "writing on the wall" and gets hung up on proving the efficacy of the drug or the validity of the theorem. From the perspective of an outsider, the ambiguity-averse DM would thus be seen to exhibit prolonged indecision.

Early on, however, the DM is concerned about the opposite scenario, namely that R is relatively unlikely and the search for evidence will be in vain. From this vantage point, her later self's refusal to discount R sufficiently enough to stop experimentation in time is seen as misguided. Recognizing the peril of continued indecision, the DM must then intervene and stop before it is too late, namely before the change in the worst-case belief occurs. The DM adopts a mixed strategy in this case, randomizing between experimentation and stopping. Mixing hedges against ambiguity, as is well-known in the literature. What is novel here is the form it takes; she randomizes in stopping according to a Poisson process at a rate suitably calibrated to hedge against both events. The stopping occurs around the point of belief switch, which is typically well before the Bayesian DM would find it reasonable to stop. Hence, to an outside observer, the DM would be seen to exhibit premature decisiveness.

Comparative statics yields further implications of ambiguity. An increase in ambiguity—as measured by the size of the initial set of priors—increases the tendency for the DM to prolong her experimentation later in the game, which increases her need to preemptively stop earlier in the process and, naturally, the rate at which she does so. These effects, taken together, imply that increased ambiguity "blunts" the DM's ability to adapt her learning strategy to the true likelihood of states. In particular, she fails to stop learning when learning becomes unprofitable (i.e., when the true likelihood of R falls low), so that given a sufficiently large ambiguity, the DM's expected experimentation time *increases* rather than decreases (as would occur under a Bayesian DM) when R becomes unlikely. The combined effects also imply that the stopping time is more dispersed with larger ambiguity.

Finally, we extend the baseline model to the case of incremental learning with diffusion signals. Although some details of the solution change, we show that the main features of the model—excessive learning and preemption via randomized stopping—persist in this case.

In sum, our analysis captures two behavioral traits—protracted indecision and hasty, often unwarranted, decisiveness. The other main contribution of the paper is methodological. Analyzing dynamic decision-making under ambiguity is a priori difficult because the associated time inconsistency problem makes the dynamic programming machinery inapplicable. We characterize the solution to our problem via a set of novel Hamilton-Jacobi-Bellman (HJB) conditions that account for potential differences in preferences of current and future selves. Our HJB characterization thereby restores the dynamic programming logic in the time-inconsistent environment and hence serves as a powerful tool kit that can make analysis tractable without compromising sharp prediction. We expect the approach, and the microfoundation justifying it, to be valuable beyond our setting.

Related Literature.—We incorporate ambiguity into an optimal stopping framework a la Wald (1947). Optimal stopping problems in the Bayesian framework with Poisson signals have been analyzed in Peskir and Shiryaev (2006, chap. VI), with economic applications including Che and Mierendorff (2019) and Nikandrova and Pancs (2018). Other applications adopt a drift-diffusion model, e.g., Moscarini and Smith (2001) and Fudenberg, Strack, and Strzalecki (2018), while Zhong (2019) studies the problem under flexible choice of information.

The current paper is part of a growing literature on optimal stopping problems under ambiguity. Most of the existing literature adopts a recursive approach, thereby precluding the possibility of changes in the worst-case scenario and time inconsistency; examples include Epstein and Schneider (2007); Riedel (2009); Miao and Wang (2011); and Cheng and Riedel (2013). Closest to our paper in this literature is Epstein and Ji (2022), which studies a Wald problem in a drift-diffusion framework with recursive preferences. Under the recursive approach, there is a single prior in the initial set that minimizes the DM's expected payoff at every point in time. Due to this feature, standard dynamic programming methods are applicable, and the optimal stopping rule continues to be described by two stopping boundaries, as in the Bayesian benchmark. By contrast, dynamic inconsistency figures prominently in our model, which we analyze by developing a saddle point version of the dynamic programming technique.

There is also a growing literature looking at the implication of dynamic inconsistency due to ambiguity in different applications. For example, Bose and Daripa (2009); Bose and Renou (2014); Ghosh and Liu (2021); and Auster and Kellner (2022) consider auctions/mechanisms with ambiguity-averse agents, while Kellner and Le Quement (2018) and Beauchêne, Li, and Li (2019) study ambiguity and updating in communication and persuasion settings.

Finally, there is a literature studying dynamic inconsistent stopping in other behavioral frameworks. For instance, Barberis (2012); Xu and Zhou (2013); Ebert and Strack (2015, 2018); and Henderson, Hobson, and Tse (2017) analyze stopping problems under prospect theory. Christensen and Lindensjö (2018, 2020) study stopping problems for a class of time-inconsistent models, which can capture endogenous habit formation, nonexponential discounting, and mean-variance utility.

I. Model

A DM must choose between two actions, r and ℓ, whose payoffs depend on an unknown state $\omega \in \{R, L\}$. Specifically, the DM's payoffs satisfy $u_r^R > u_\ell^R$ and $u_\ell^L > u_r^L$, where $u_a^\omega \in \mathbb{R}$ denotes her payoff from choosing action $a \in \{r, \ell\}$ in state $\omega \in \{R, L\}$. The DM thus wants to "match" the action with the state. We further assume $u_r^R > u_r^L$ and $u_\ell^R < u_\ell^L$. This means that whether a state is desirable or not depends on the action that the DM takes, a feature that makes ambiguity nontrivial.[1] Let $U_a(p) = p u_a^R + (1-p) u_a^L$ denote the DM's payoff associated with action a when the probability of state R is p.

The DM may delay her action and acquire information. We model the DM's information acquisition problem as a stopping problem in continuous time. At each point in time $t \geq 0$, the DM decides whether to wait for more information or whether to stop and take an immediate action, ℓ or r. If the DM waits (or "experiments"), she incurs a flow cost $c > 0$ per unit of time. Payoffs are realized when the DM stops and takes an action. Information takes the form of R-evidence, which is received only in state R with arrival rate $\lambda > 0$. Observing news thus conclusively reveals state R, while the absence of news is a signal favoring state L.

Bayesian Benchmark.—Considering the benchmark case of a Bayesian DM, let p_t denote the probability that the DM assigns to state R at time t. In the absence of news, the DM's belief drifts "leftward" according to the law of motion:

$$\dot{p}_t = \eta(p_t) := -\lambda p_t (1 - p_t). \tag{1}$$

Intuitively, the DM becomes pessimistic toward R when she looks for R-evidence but is unable to find any.

As is well-known, the Bayesian optimal information acquisition rule can be described by two thresholds, p_ℓ^B and p_r^B. If $p_t \leq p_\ell^B$ or $p_t \geq p_r^B$, the DM is sufficiently confident in the state and takes the appropriate action without gathering any

[1] If this assumption does not hold, the ambiguity-averse DM minimizing over a set of probabilities is behaviorally indistinguishable from a Bayesian DM, whose prior is equal to the element of the set that maximizes the likelihood of the "bad" state.

additional information. If instead p_t belongs to $\left(p_\ell^B, p_r^B\right)$, the DM experiments until either she receives R-evidence or the Bayesian update of p_t reaches the threshold p_ℓ^B.

$$0 \underbrace{\hspace{5em}}_{\text{action } \ell} p_\ell^B \underbrace{\leftarrow\leftarrow\leftarrow\leftarrow\leftarrow\leftarrow\leftarrow\leftarrow}_{\text{experimentation}} p_r^B \underbrace{\hspace{5em}}_{\text{action } r} 1.$$

Let $\phi(p; p', U_\ell(p'))$ denote the value to the Bayesian DM of experimenting until the belief p reaches $p' < p$, assuming that if she receives R-evidence during the experimentation, she realizes u_r^R, while if she does not receive such evidence until her belief reaches p', she stops and takes action ℓ. The standard HJB conditions characterize the DM's value function as a solution to an ODE:

$$c = \lambda p\left[u_r^R - \phi(p)\right] + \phi'(p)\eta(p), \ \forall p > p'. \tag{2}$$

Intuitively, the flow cost of experimentation (LHS) equals the rate of value increase due to possible breakthrough news and belief updating in its absence (RHS). Optimal stopping calls for p' to be chosen optimally, so the value of the stopping problem is

$$\Phi(p) = \max_{p' \in [0,p]} \phi(p; p', U_\ell(p')), \tag{3}$$

and the left stopping boundary p_ℓ^B is the associated maximizer.[2] Finally, since the DM has the option to choose r at any point in time, the DM with belief p enjoys a payoff of

$$\Phi^*(p) = \max\left\{\Phi(p), U_r(p)\right\}, \tag{4}$$

with the DM indifferent between experimentation and action r at the right stopping boundary p_r^B. We will call $\Phi^*(p)$ the *Bayesian value function* throughout. Its characterization is relegated to online Appendix B.1.

Figure 1 illustrates graphically the value of the Bayesian DM in a typical situation. The downward-sloping line shows the DM's expected payoff from taking action ℓ as a function of the probability p she assigns to state R. Similarly, the upward-sloping line represents the DM's expected payoff from taking action r. The crossing point $\hat{p}$ is the belief at which the DM is indifferent between actions ℓ and r. The blue curve depicts Φ, i.e., the value from experimenting with stopping belief p_ℓ^B. The optimal value Φ^* can be seen as the upper envelope of Φ and $U := U_\ell \vee U_r$. Naturally, the experimentation range $\left(p_\ell^B, p_r^B\right)$ depends on the flow cost c; the range shrinks as c increases and disappears when

$$c \geq \bar{c} := \lambda \frac{\left(u_r^R - u_\ell^R\right)\left(u_\ell^L - u_r^L\right)}{\left(u_r^R - u_\ell^R\right) + \left(u_\ell^L - u_r^L\right)}. \tag{5}$$

The threshold $\bar{c}$ is the value of c at which $\Phi(\hat{p}) = U(\hat{p})$.

[2] One can characterize the optimal stopping boundary p_ℓ^B by invoking value matching and smooth pasting. Note also that if c is sufficiently large, then the optimal solution is to stop immediately; that is, $p' = p$.

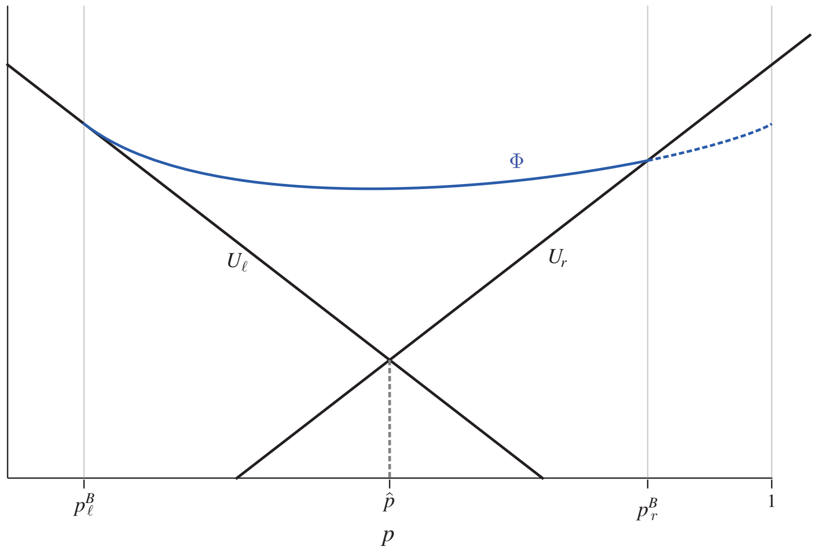

FIGURE 1. BAYESIAN OPTIMAL STOPPING RULE

Ambiguity and Updating.—We assume that the DM faces ambiguity over the likelihood of state R and that her preferences are represented by the maxmin expected utility model (Gilboa and Schmeidler 1989). According to this model, the DM seeks to maximize her payoff guarantee across a set of priors, described by an interval of probabilities on state R: $\mathcal{P}_0 = [\underline{p}_0, \overline{p}_0]$. We assume $0 < \underline{p}_0 \leq \overline{p}_0 < 1$.

A key question is how the DM updates her ambiguous beliefs as information arrives. We assume that the DM uses *Full Bayesian Updating*: she updates each element of $\mathcal{P}_0$ according to Bayes' rule and considers the worst case over the set of posterior beliefs obtained in this way. The set of posteriors at time $t > 0$ is thus given by an interval $\mathcal{P}_t := [\underline{p}_t, \overline{p}_t]$, where $\underline{p}_t$ is the Bayesian update of the initial lower bound $\underline{p}_0$ and $\overline{p}_t$ is the Bayesian update of the initial upper bound $\overline{p}_0$.

As time progresses and no breakthrough occurs, the interval $\mathcal{P}_t$ drifts to the left. In the process, the interval expands in size as it moves away from the right boundary and shrinks as it approaches the left boundary (ultimately, both $\underline{p}_t$ and $\overline{p}_t$ converge to zero). Rescaling the belief in terms of a log-likelihood ratio keeps the interval "size" constant. Hence, from now on, we shall refer to

$$\Delta := \ln\left(\frac{\overline{p}_0}{1 - \overline{p}_0}\right) - \ln\left(\frac{\underline{p}_0}{1 - \underline{p}_0}\right)$$

as *the degree of ambiguity*. It is easy to verify that Δ does not vary as one updates beliefs over time.

Preference Reversals.—As is well-known, the Full Bayesian Updating rule can lead to violations of dynamic consistency.[3] This feature will play a crucial role in

[3] Beyond the issue of dynamic inconsistency, Full Bayesian Updating, also referred to as *prior-by-prior updating*, has been criticized in the literature for implying a possible situation in which "all news is bad news" (see Gul

our model. To explain the source of dynamic inconsistency in our setting, it will be useful to first consider the DM's optimal information acquisition rule from the perspective of time 0; we will then show why, after updating the belief set, the DM may have the incentive to deviate from it.

The ex ante optimal rule solves the DM's optimization problem in a hypothetical setting where she can commit to a dynamic strategy that maximizes her expected payoff against the worst prior in $\mathcal{P}_0$. This optimization problem is clearly independent of the assumed updating rule and thus constitutes an important benchmark. Finding the solution amounts to analyzing a zero-sum game between the DM and adversarial nature choosing $p \in \mathcal{P}_0$ to minimize the DM's expected payoff. Its equilibrium—a saddle point—describes the DM's maxmin optimal value obtained from any information acquisition rule. Online Appendix B.2 establishes the existence of a saddle point. Given a saddle point, its value also equals the value of the minmax problem in which nature first chooses a prior in $\mathcal{P}_0$ and the DM then selects the best strategy for the chosen prior (see Osborne and Rubinstein 1994, Proposition 22.2, for example). It follows that the minmax value is simply the lowest Bayesian value Φ^* within $\mathcal{P}_0$, or

$$\min_{p \in \mathcal{P}_0} \Phi^*(p).$$

This value coincides with the Bayesian value at the leftmost belief $\underline{p}_0$ if $\mathcal{P}_0$ is sufficiently far to the right, at the rightmost belief $\bar{p}_0$ if $\mathcal{P}_0$ is sufficiently far to the left, and at an interior belief if $\mathcal{P}_0$ contains the global minimizer of Φ^* (see Figure 2 for the first two cases). For later purposes, we denote the minimizer of Φ^* by

$$p_* := \arg\min_{p \in [0,1]} \Phi^*(p).$$

While the maxmin commitment value is easily obtained, the saddle point strategy on the part of the DM requires a little care. As we show in online Appendix B.2, if $p_* < p_r^B$ or $p_r^B \notin \mathcal{P}_0$, then the maxmin strategy is indeed the Bayesian optimal strategy for the worst belief in $\mathcal{P}_0$. If $p_* = p_r^B \in \mathcal{P}_0$, however, nature's indifference, which is needed for the saddle point requirement, calls for the DM to randomize. If $c < \bar{c}$, then the DM must randomize between r and experimentation. If instead $c \geq \bar{c}$, experimentation is too costly, and the DM mixes between actions r and ℓ with probabilities $\hat{\rho}$ and $1 - \hat{\rho}$, respectively, where $\hat{\rho}$ satisfies

$$\hat{\rho}\, U_r'(p) + (1 - \hat{\rho}) U_\ell'(p) = 0. \tag{6}$$

To see now why the described solution may be incompatible with the preferences of the DM's later selves, suppose the interval $\mathcal{P}_0$ is sufficiently far to the right so that $\underline{p}_0$ minimizes Φ^* over $\mathcal{P}_0$ (see the left panel of Figure 2). If $\underline{p}_0 < p_r^B$, the DM's optimal (commitment) plan at time 0 is to experiment until either she receives R-evidence or the Bayesian update $\underline{p}_t$ of $\underline{p}_0$ reaches p_ℓ^B. Once the intended stopping point t, where $\underline{p}_t = p_\ell^B$, is reached, however, the worst-case belief is no longer

and Pesendorfer 2021 and Shishkin and Ortoleva 2023). Since in our setting R-evidence is always good news, this type of situation does not arise here.

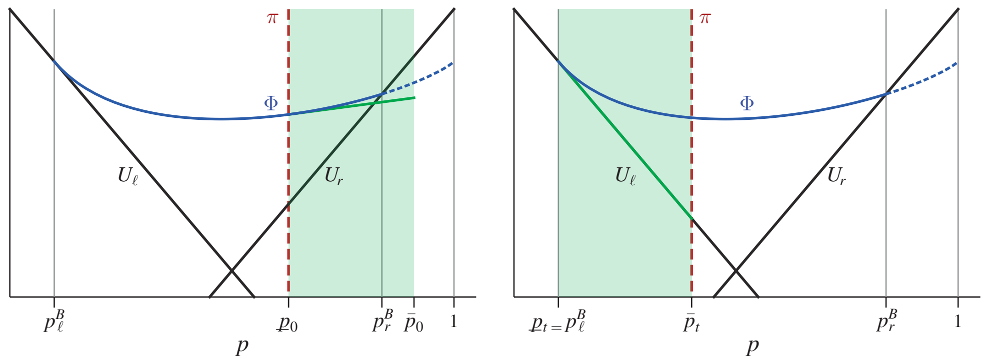

Figure 2. Worst-Case Scenario (Indicated by the Red Dashed Line) at the Start of Experimentation (Left Panel) and at the Ex Ante Optimal Stopping Time (Right Panel)

given by $\underline{p}_t$ but instead by $\bar{p}_t$, as illustrated in the right panel of Figure 2. According to $\bar{p}_t$, the optimal plan (as prescribed by the commitment solution) is to acquire more information, until the Bayesian update of $\bar{p}_t$ reaches the stopping boundary p_ℓ^B. Hence, later selves of the DM will renege on the original plan prescribed by the commitment solution and *experiment excessively* from the perspective of the original self.

A dynamic inconsistency arises as a result of a switch in the worst-case scenario. When the DM experiments, she is concerned about two events: (i) not receiving R-evidence in state L and (ii) taking action ℓ in state R. At the start of the experimentation phase, pessimism about the former event dominates, and the worst-case belief is given by $\underline{p}_t$, which minimizes the likelihood of R. When the DM gets closer to the intended stopping point, however, the concern about taking the wrong action looms large, and the worst-case belief switches to $\bar{p}_t$, i.e., the belief maximizing the likelihood of R. The new worst-case belief calls for more experimentation, thereby creating a conflict with the preferences of earlier selves. A sophisticated DM understands that later selves deviate from the commitment plan by experimenting for too long and optimizes accordingly. We next analyze the behavior of such a DM.

II. Sophisticated Stopping Rule

We now analyze the dynamic behavior of a sophisticated DM. The dynamic inconsistency problem requires us to analyze the DM's stopping problem as an intrapersonal game in which a current self plays against future selves.

A. *Intrapersonal Game*

At each point in time $t \geq 0$, the DM decides between continuing to gather information and stopping, taking as given the choices of her future selves. If a breakthrough occurs, all posteriors in the updated set are equal to one, so it is optimal for the DM to stop and take action r. Taking this part of the strategy as given, it is without loss of generality to focus on the Markov strategy that depends on the set of posteriors rather than time. Indeed, in the absence of R-evidence, there is a one-to-one mapping between calendar time and the set of posteriors. Having fixed Δ as a primitive, we can index the possible sets of posteriors by their upper bound $\bar{p} \in (0,1)$. The rightmost belief in the set, $\bar{p}$, is thus our state variable. We denote the set of posteriors at state $\bar{p}$ by

$$\mathcal{P}(\bar{p}) \coloneqq \left\{p \leq \bar{p} : \ln\left(\frac{\bar{p}}{1-\bar{p}}\right) - \ln\left(\frac{p}{1-p}\right) \leq \Delta\right\},$$

and define $\underline{p}(\bar{p}) \coloneqq \min \mathcal{P}(\bar{p})$ as the lower bound of this set. Clearly, it follows that $\underline{p}_t = \underline{p}(\bar{p}_t)$.

A Markov strategy is a measurable function $\sigma : (0,1) \to [0,\infty) \times [0,1] \times [0,1]$, with $\sigma(\bar{p}) = (\nu(\bar{p}), m(\bar{p}), \rho(\bar{p}))$. The strategy specifies a *stopping rate* $\nu(\bar{p})$, an *instantaneous stopping probability* $m(\bar{p})$, and the *probability of taking action r conditional on stopping* $\rho(\bar{p})$. Intuitively, densities in the distribution of stopping times are accommodated by interior stopping rates $\nu(\bar{p})$, while atoms in the stopping-time distribution are captured by positive values of $m(\bar{p})$.[4] To ensure that a strategy σ induces a well-defined stopping time, we impose the following *admissibility* restrictions on strategies: (i) every connected set $\mathcal{P} \subseteq [0,1]$ with $m(\bar{p}) = 1$ for all $\bar{p} \in \mathcal{P}$ contains its supremum; (ii) there is a countable set $M^\sigma \subset (0,1)$ such that $m(\bar{p}) \in (0,1)$ only if $\bar{p} \in M^\sigma$.[5]

Fixing a strategy σ, we denote by $V^\sigma(p,\bar{p})$ the DM's expected payoff at state $\bar{p}$ when the strategy is evaluated at belief p. The detailed expression of $V^\sigma(p,\bar{p})$ can be found in Appendix A1; see equation (A3).

Nature's goal is to minimize $V^\sigma(p,\bar{p})$ at each state $\bar{p}$ by choosing $p \in \mathcal{P}(\bar{p})$. Since $V^\sigma(p,\bar{p})$ is a convex combination of the payoffs that strategy σ yields in state R and in state L with coefficient p, the DM's value $V^\sigma(p,\bar{p})$ is linear in the first argument. Nature's problem thus has a corner solution, except when $V_p^\sigma(\cdot,\bar{p}) = 0$. Letting $\pi(\bar{p}) \in \mathcal{P}(\bar{p})$ denote nature's choice at state $\bar{p}$, we define a solution concept as follows.

DEFINITION 1: *A strategy $\sigma = (\nu, m, \rho)$ constitutes a pseudo equilibrium if for each state $\bar{p} \in (0,1)$, there exists some $\pi = \pi(\bar{p})$ with*

$$\pi(\bar{p}) \in \underset{p \in \mathcal{P}(\bar{p})}{\arg\min}\, V^\sigma(p,\bar{p}), \tag{7}$$

such that[6]

$$U(\pi) \leq V^\sigma(\pi,\bar{p}); \tag{8}$$

$$U(\pi) = V^\sigma(\pi,\bar{p}) \quad \textit{if} \quad \nu(\bar{p}) \quad \textit{or} \quad m(\bar{p}) > 0; \tag{9}$$

[4] One can show that our Markov strategy is equivalent to an alternative formulation that specifies the DM's strategy by a family of stopping times, or more precisely, by a family $\{F_t\}_{t\geq 0}$ of distributions satisfying some restrictions such as right-differentiability, where $F_t(\tau)$ is the probability that the DM stops by time $t + \tau$ conditional on not having received any breakthrough news by time t. See online Appendix B.3 for details.

[5] These conditions correspond to the requirement that the stopping-time distribution is right-continuous and differentiable at the points of continuities. Specifically, condition (i) ensures that the stopping time is well defined when the belief drifts toward a (left) stopping boundary. Meanwhile, condition (ii) corresponds to the fact that a distribution function can have at most countably many points of discontinuities.

[6] Recall that $U = U_\ell \vee U_r$ is the maximum stopping payoff from actions ℓ and r.

$$\rho(\bar{p}) \in \underset{\rho \in [0,1]}{\arg\max}\, \rho U_r(\pi) + (1-\rho) U_\ell(\pi). \tag{10}$$

A strategy σ is a pseudo equilibrium if, given nature's choice π, no self of the DM has the incentive to deviate.[7] Due to the continuous-time nature of our game, many pseudo equilibria exist. For instance, it is a pseudo equilibrium for the DM to stop at every state and choose the optimal action against the worst-case belief. If some self of the DM were to deviate and refuse to stop, her infinitesimally close future selves would stop immediately. Since experimentation for an instant yields a zero probability of a breakthrough, the deviation is never profitable. However, the prescribed strategy is implausible and simply an artifact of the continuous-time nature of our game.

To rule out such implausible strategies, we focus on the pseudo equilibria in which the DM can control her action for a vanishingly small amount of time. Formally, for any pseudo equilibrium σ, we imagine a DM who can commit at each time t her action for the period $[t, t+\epsilon)$ against adversarial nature, anticipating that σ will follow after $t+\epsilon$. We then require σ to be obtained as a limit of such ϵ-commitment solutions as $\epsilon \to 0$.[8] Such a vanishing control over future behavior avoids total coordination failure among the different selves. As justified in Appendix A2 more precisely, the refinement introduces a version of dynamic programming conditions.

To begin, for a value function $V : [0,1]^2 \to \mathbb{R}$, define a functional

$$\begin{aligned} G(m,\nu,\rho,p,\bar{p},V,dV) = {} & m\big[U_\rho(p) - V(p,\bar{p}_-)\big] + (1-m) \\ & \times \Big[-c + \nu\big[U_\rho(p) - V(p,\bar{p}_-)\big] + p\lambda\big[u_r^R - V(p,\bar{p}_-)\big] \\ & + V_p(p,p_-)\eta(p) + V_{\bar{p}}(p,\bar{p}_-)\eta(\bar{p})\Big], \end{aligned}$$

where dV is the gradient of the value function, $V(p,\bar{p}_-) := \lim_{\bar{p}'\uparrow\bar{p}} V(p,\bar{p}')$, and $V_x(p,\bar{p}_-) := \lim_{\bar{p}'\uparrow\bar{p}} \partial V(p,\bar{p}')/\partial x$.[9] Recall $\eta(p) = -\lambda p(1-p)$ denotes the law of motion. The functional G has a familiar interpretation from the dynamic programming literature. Its first term captures a value increase when the DM stops with a point mass m; she collects $U_\rho(p) - V(p,\bar{p}_-)$. The second term (inside the square brackets) captures a value increase when the DM "continues." In this case, at the flow cost c, the DM engages in a flow stopping at rate ν and experimentation yielding a breakthrough at rate λp: her value increases by $U_\rho(p) - V(p,\bar{p}_-)$ in the former case and by $u_r^R - V(p,\bar{p}_-)$ in the latter case. The last two terms track

[7] Note that the conditions in Definition 1 and later in Definition 2 make no reference to the initial state $\bar{p}_0$. This means that the same strategy should be employed, *regardless of the initial state.* While this is, in principle, a restriction, this restriction entails no loss except for one knife-edge case. See footnote 15 for details.

[8] This refinement is similar in effect to taking a limit of the behavior in discrete times models as the time interval shrinks to zero; in such models, the DM's experimentation during a unit period has a nonnegligible probability of producing a breakthrough. The current refinement accomplishes the same effect within a continuous-time framework.

[9] We use the left limits of the value function and its partial derivatives (i) to make sure the conditions are well defined even at points where the derivatives do not exist and (ii) since the belief drifts downward and the value function in the HJB equation reflects the continuation value.

the change of value arising from the updating of the state $\bar{p}$ and the updating of the current worst-case belief. This part of the HJB is nonstandard and accounts for dynamically inconsistent belief changes. To compare, in the standard case without ambiguity, the DM's belief p is the only payoff-relevant variable. Hence, the first term $V_p^{\sigma}(p,\bar{p})$ would be present, while the second-term $V_{\bar{p}}^{\sigma}(p,\bar{p})$ would not. In the current case, the worst-case belief tomorrow may differ from tomorrow's update of today's worst-case belief,[10] making it necessary to account for changes both in the state $\bar{p}$, which determines the continuation strategy, and in the belief p with which the strategy is evaluated.

For each $\bar{p} \in (0,1)$, consider a saddle point version of HJB equations:[11]

$$(11) \qquad 0 = G\big(m(\bar{p}),\nu(\bar{p}),\rho(\bar{p}),\pi(\bar{p}),\bar{p},V,dV\big),$$

$$(12) \qquad \sigma(\bar{p}) = \big(m(\bar{p}),\nu(\bar{p}),\rho(\bar{p})\big) \in \underset{m,\nu,\rho}{\arg\max}\, G\big(m,\nu,\rho,\pi(\bar{p}),\bar{p},V,dV\big),$$

$$(13) \qquad \pi(\bar{p}) \in \underset{p\in\mathcal{P}(\bar{p})}{\arg\min}\, G\big(m(\bar{p}),\nu(\bar{p}),\rho(\bar{p}),p,\bar{p},V,dV\big).$$

Condition (11) captures the optimality requirement embodied in the value function: the variation from the optimal value for the DM is precisely zero at the equilibrium. Condition (12) requires the DM's flow decision to be optimal against nature's choice π, taking her future strategy as given. This is not required in Definition 1, as is seen in the implausible "always stopping" behavior. Finally, equation (13) chooses the belief to be worst for the DM against the flow action, taking the continuation strategy into consideration. Effectively, equations (11), (12), and (13) represent the saddle point version of HJB equations. We define our equilibrium as follows.

DEFINITION 2: *A pseudo equilibrium $\sigma = (\nu,m,\rho)$, together with π, is an intrapersonal equilibrium*, *or simply an equilibrium*, *if it satisfies equations* (11), (12), *and* (13) *for each $\bar{p} \in (0,1)$.*

The precise microfoundation for this solution concept based on vanishing commitment power is provided in Appendix A2. It is important to note that the purpose of our HJB equations differs from the standard one, which is to characterize the long-term optimal strategy. Recall that our DM cannot control her future selves, so her strategy is typically not long-term optimal (even given her adversarial belief). Instead, our HJB equations characterize the (vanishingly) near-term maxmin-optimality of her strategy, as explained above, thereby refining the set of potential equilibria. This use of HJB conditions for dynamic decisions under ambiguity, and the microfoundation justifying them, are of independent interest and are applicable beyond the particular setting we consider here. While time inconsistency

[10] Specifically, a dynamic inconsistency arises when $\pi\big[\bar{p} + \eta(\bar{p})dt\big] \neq \pi(\bar{p}) + \eta\big[\pi(\bar{p})\big]dt$.

[11] Similar to the commitment solution, a saddle point here characterizes the maxmin optimality of the DM's strategy (for a vanishingly near future). Equations (11)–(13) resemble Isaac's condition, which characterizes the equilibrium of dynamic zero-sum games played by two long-run players (see Bardi and Capuzzo-Dolcetta 1997, p. 446 and Laraki and Solan 2004). The fact that our DM is not a time-consistent long-run player differentiates our condition in its function and purpose from Isaac's condition.

makes the dynamic programming machinery inapplicable, the HJB characterization restores the dynamic programming logic in a saddle point and near-term optimality sense. This characterization serves as a powerful tool kit that can make analysis tractable without compromising sharp prediction, as we argue is the case here.

B. *Main Result*

To state our main result, we introduce an additional threshold for the flow cost c:

$$\underline{c} := \frac{\delta_r}{\delta_r + \delta_\ell}\bar{c},$$

where $\delta_a = |u_a^R - u_a^L|$ denotes the payoff difference across the two states given the DM's action $a \in \{\ell, r\}$. As can be easily seen, $\underline{c}$ is strictly smaller than $\bar{c}$, i.e., the Bayesian threshold defined in Section I. We will focus on the case in which the Bayesian value function Φ^* attains its minimum at $p_* < p_r^B$ such that $\Phi'(p_*) = 0$, as graphed in Figure 3. This case, labeled Case 1, occurs when the value of outright action r is not too high. The opposite case, labeled Case 2, in which $p_* = p_r^B$, or $\Phi'(p_*) < 0$ (as graphed in Figure 7), will be treated at the end of this section. This is also where we discuss the conditions under which the equilibrium we describe is unique (and under which it is not).

THEOREM 1: *The following intrapersonal equilibrium exists.*

(*i*) *Suppose $c \geq \bar{c}$. The DM does not acquire information. If $\bar{p} \leq \hat{p}$, she takes action ℓ; if $\underline{p}(\bar{p}) \geq \hat{p}$, she takes action r; if $\hat{p} \in (\underline{p}(\bar{p}), \bar{p})$, she randomizes between ℓ and r with probability $\hat{\rho}$.*

(*ii*) *Suppose $c < \bar{c}$ and $\Phi'(p_*) = 0$. For each $c \in (0, \bar{c})$, there is a threshold $\Delta_c \in (0, +\infty]$ with $\Delta_c = +\infty$ if and only if $c \leq \underline{c}$ such that the intrapersonal equilibrium is described as follows*:

(*a*) *If $\Delta \leq \Delta_c$, there exist cutoffs $0 < \bar{p}_1 < \bar{p}_2 \leq \bar{p}_3 < \bar{p}_4 < 1$ with $\bar{p}_1 = p_\ell^B$ and $\bar{p}_2 = p_*$ such that*[12]

$$\big(m(\bar{p}), \nu(\bar{p}), \rho(\bar{p})\big) = \begin{cases} (1,0,0); \\ (0,0,\cdot); \\ (0,\nu^*(\bar{p}),0); \\ (0,0,\cdot); \\ (1,0,1); \end{cases} \qquad \pi(\bar{p}) = \begin{cases} \bar{p}, & \text{if } \bar{p} \in [0,\bar{p}_1]; \\ \bar{p}, & \text{if } \bar{p} \in (\bar{p}_1,\bar{p}_2); \\ \pi^*(\bar{p}), & \text{if } \bar{p} \in (\bar{p}_2,\bar{p}_3); \\ \underline{p}(\bar{p}), & \text{if } \bar{p} \in [\bar{p}_3,\bar{p}_4); \\ \underline{p}(\bar{p}), & \text{if } \bar{p} \in [\bar{p}_4,1]; \end{cases}$$

[12] We leave ρ unspecified if $m = \nu = 0$.

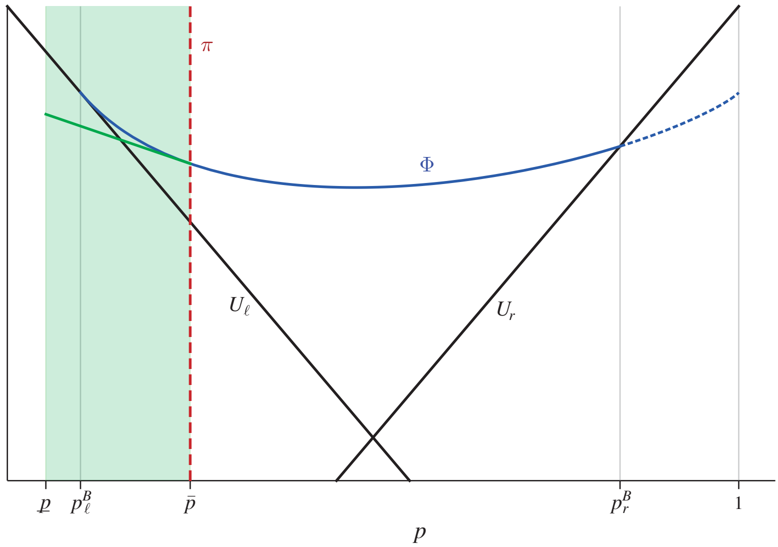

Figure 3. The Value Segment (Green) and Worst-Case Belief for $\bar{p} \in (\bar{p}_1, \bar{p}_2)$

where $\nu^*(\bar{p}) > 0$ *and* $\pi^*(\bar{p}) \in (\underline{p}(\bar{p}), \bar{p})$, *for* $\bar{p} \in (\bar{p}_2, \bar{p}_3)$, *and* $\bar{p}_2 < \bar{p}_3$ *if and only if* $\Phi(p^*) < U_\ell(\underline{p}(p^*))$.

(*b*) *If* $\Delta > \Delta_c$, *the equilibrium has the same structure as in* (*a*) *except that*

$$\big(m(\bar{p}), \nu(\bar{p}), \rho(\bar{p})\big) = (1, 0, \hat{\rho}) \quad \textit{and} \quad \pi(\bar{p}) = \hat{p}$$

for $\bar{p} \in [\bar{p}_3, \bar{p}_4)$.

We structure the discussion of Theorem 1 according to the level of experimentation costs. If $c \geq \bar{c}$, the DM does not experiment, and her equilibrium strategy coincides with the commitment solution described in Proposition 2. This is a consequence of the facts that (i) given $c \geq \bar{c}$, the DM's commitment solution involves no experimentation, as we have seen in Section I, and (ii) implementing this solution requires no commitment. The DM's value in this case is $V^\sigma(\pi(\bar{p}), \bar{p}) = \min_{p \in \mathcal{P}(\bar{p})} U(p)$.

Our main focus will lie on the case $c \leq \underline{c}$. Under this restriction, the equilibrium is always described by Theorem 1(ii)(a) and can be illustrated as follows:

$$0 \underbrace{\qquad\qquad}_{\text{action } \ell} \bar{p}_1 \underbrace{\leftarrow\leftarrow\leftarrow\leftarrow}_{\text{pure exp.}} \bar{p}_2 \underbrace{\leftarrow\leftarrow\leftarrow\leftarrow\leftarrow}_{\text{mixed stopping } (\ell)} \bar{p}_3 \underbrace{\leftarrow\leftarrow\leftarrow\leftarrow}_{\text{pure exp.}} \bar{p}_4 \underbrace{\qquad\qquad}_{\text{action } r} 1.$$

When $\bar{p}$ is sufficiently extreme ($\bar{p} \leq \bar{p}_1$ or $\bar{p} \geq \bar{p}_4$), the DM takes immediate action without gathering additional information, just like the Bayesian DM. For the remaining states, however, the characterization differs qualitatively from the benchmark case. In particular, if Δ is sufficiently large, there is a middle region of states where the DM randomizes between experimentation and action ℓ at a positive rate. As a result, the region of pure experimentation is no longer convex.

To derive the solution, we leverage the fact that in the absence of breakthrough news, the set of beliefs drifts to the left. Hence, we explain the logic of the equilibrium strategy via backward induction, starting from a far left set of beliefs.

Region 1: $\bar{p} \leq \bar{p}_1$. The first threshold $\bar{p}_1$ coincides with the Bayesian stopping boundary p_ℓ^B. If $\bar{p} \leq p_\ell^B$, then stopping followed by action ℓ is optimal for all beliefs in $\left[\underline{p}(\bar{p}), \bar{p}\right]$ and thus constitutes the equilibrium strategy for our ambiguity-averse DM. The worst-case belief under this strategy is $\bar{p}$.

Region 2: $\bar{p} \in (\bar{p}_1, \bar{p}_2]$. In Region 2, the rightmost belief $\bar{p}$ continues to be the worst-case belief for all $\bar{p} \in (\bar{p}_1, \bar{p}_2]$, and the DM follows the Bayesian optimal stopping rule with respect to $\bar{p}$. Since $\bar{p} > \bar{p}_1 = p_\ell^B$, this strategy entails experimentation until either a breakthrough occurs or the Bayesian update of $\bar{p}$ reaches p_ℓ^B. The right boundary of this region is $\bar{p}_2 := p_*$, the belief at which the Bayesian value function attains its minimum.

Fix any state $\bar{p} \in (\bar{p}_1, \bar{p}_2]$. We can evaluate the expected payoff $V^\sigma(p, \bar{p})$ of strategy σ as a function of the belief p. The set of values corresponding to possible alternative beliefs, $\left\{\left(p, V^\sigma(p, \bar{p})\right) : p \in \mathcal{P}(\bar{p})\right\}$, forms a linear segment, which we will call *value segment* throughout (see, for instance, Figure 3). It is useful to track the movement of the value segment, as time moves backward or forward. Note that the value segment is tangent to the value function Φ at $\bar{p}$, due to the fact that the stopping rule is Bayesian optimal with respect to $\bar{p}$.[13] Tangency at this point implies that the value segment is downward sloping and, hence, that $V^\sigma(p, \bar{p})$ is indeed minimized at $p = \bar{p}$. Hence, $\pi(\bar{p}) = \bar{p}$ is nature's optimal choice against $\sigma(\bar{p})$, and the prescribed $\sigma(\bar{p})$, being Bayesian optimal for $p = \bar{p}$, is a best response against $\pi(\bar{p})$. The profile $\left(\sigma(\bar{p}), \pi(\bar{p})\right)$ thus forms a saddle point for state $\bar{p}$.

At $\bar{p} = p_* =: \bar{p}_2$, the value segment becomes flat (see Figure 4), as we assumed $\Phi'(p_*) = 0$, so all posteriors in $\left[\underline{p}(\bar{p}), \bar{p}\right]$ minimize the DM's expected payoff.

Region 3: $\bar{p} \in (\bar{p}_2, \bar{p}_3]$. Moving further back in time, suppose the state $\bar{p}$ is slightly higher than p_* and the DM contemplates experimenting according to the strategy prescribed in Region 2. The value segment becomes now upward sloping, implying that the worst-case scenario is described by $\underline{p}(\bar{p})$ rather than $\bar{p}$. We then distinguish two cases. If the degree of ambiguity Δ is sufficiently small and, hence, the value segment is sufficiently "short," experimentation is optimal at the worst-case belief $\underline{p}(\bar{p})$ (see Figure 4, left panel).[14] The randomization region, Region 3, is empty here, and we directly transition to Region 4 at $\bar{p} = p_*$.

More interestingly, suppose Δ is sufficiently large such that the value segment crosses the stopping payoff line U_ℓ from below. Then, at state $\bar{p} = p_*$ experimentation is dominated by stopping followed by ℓ according to the worst-case belief $\underline{p}(p_*)$

[13] Bayesian optimality of the strategy with respect to $\bar{p}$ clearly implies $V^\sigma(\bar{p}, \bar{p}) = \Phi(\bar{p})$. This, together with linearity of $V^\sigma(p, \bar{p})$ in the first argument and the fact that $\Phi(p) \geq V^\sigma(p, \bar{p})$ holds for all $p \in (p_\ell^B, p_r^B)$, implies that the value segment is tangent to Φ at $\bar{p}$.

[14] Formally, we have $V^\sigma\left(\underline{p}(p_*), p_*\right) \geq U_\ell\left(\underline{p}(p_*)\right)$.

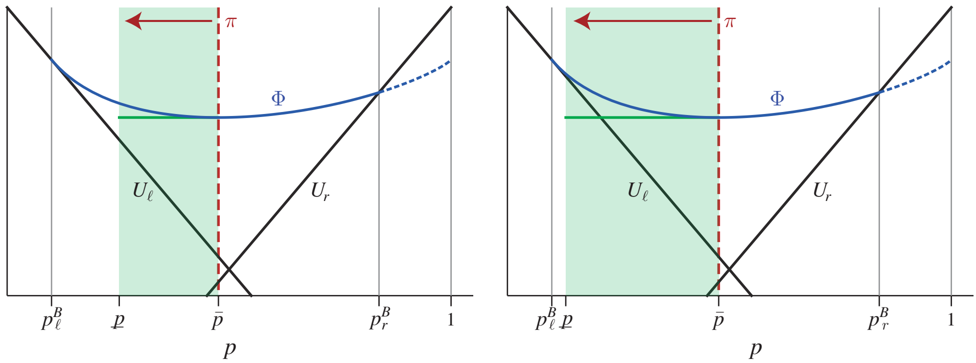

FIGURE 4. THE VALUE SEGMENT AND WORST-CASE BELIEF FOR $\bar{p} = \bar{p}_2$ WITH $V^\sigma\big(\underline{p}(p_*),p_*\big) \geq U_\ell\big(\underline{p}(p_*)\big)$ (LEFT PANEL) AND $V^\sigma\big(\underline{p}(p_*),p_*\big) < U_\ell\big(\underline{p}(p_*)\big)$ (RIGHT PANEL)

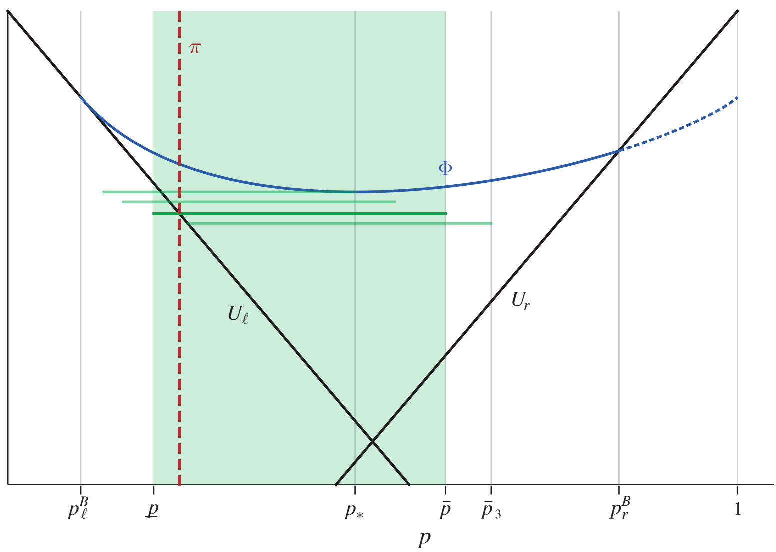

FIGURE 5. THE VALUE SEGMENT AND WORST-CASE BELIEF FOR $\bar{p} \in (\bar{p}_2,\bar{p}_3)$

(see Figure 4, right panel). Choosing ℓ, however, cannot be an equilibrium either, as nature would now choose $\bar{p}$, which makes it optimal for the DM to experiment. The intrapersonal equilibrium is thus in mixed strategies, where the DM randomizes between experimentation and action ℓ.

The DM's value and the equilibrium stopping rate in the mixed stopping region can be derived via our HJB conditions. Randomization between experimentation and action ℓ requires that the coefficient of ν in (12) must vanish, and thus,

$$V\big(\pi(\bar{p}),\bar{p}\big) = U_\ell\big(\pi(\bar{p})\big). \tag{14}$$

Graphically, the stopping payoff U_ℓ and the value segment must cross at the worst-case belief $\pi(\bar{p})$, making the DM indifferent between experimentation and action ℓ. Since the required belief $\pi(\bar{p})$ will be in the interior of $\left[\underline{p}(\bar{p}),\bar{p}\right]$, for nature to optimally choose it, the value segment must remain flat so that all posteriors in $\left[\underline{p}(\bar{p}),\bar{p}\right]$ minimize the DM's expected payoff. We can thus write $V(p,\bar{p}) = \hat{V}(\bar{p}), \forall p$. Substituting this into the DM's indifference condition (14), we obtain nature's choice

$$\pi(\bar{p}) = \frac{u_\ell^L - \hat{V}(\bar{p})}{u_\ell^L - u_\ell^R}. \tag{15}$$

The randomization by the DM in turn implies that the derivative of the objective in equation (12) with respect to p must vanish. This fact, together with $V(p,\bar{p}) = \hat{V}(\bar{p}), \forall p$, yields the equilibrium stopping rate

$$\nu(\bar{p}) = \frac{\lambda\left[u_r^R - \hat{V}(\bar{p})\right]}{u_\ell^L - u_\ell^R}. \tag{16}$$

Using equations (15) and (16), the value function $\hat{V}(\bar{p})$ is derived by substituting for π and ν in the HJB condition (11), which gives rise to the following differential equation:

$$\bar{p}(1-\bar{p})\hat{V}'(\bar{p}) = \frac{\left[u_r^R - \hat{V}(\bar{p})\right]\left[u_\ell^L - \hat{V}(\bar{p})\right]}{u_\ell^L - u_\ell^R} - \frac{c}{\lambda}. \tag{$\widehat{\text{ODE}}$}$$

Together with the boundary condition $\hat{V}(\bar{p}_2) = \Phi(\bar{p}_2)$, ($\widehat{\text{ODE}}$) admits a unique solution, describing the DM's value in Region 3.

The reason that the DM stops according to a Poisson process, and thus with vanishing probability (rather than positive probability), is because when $\bar{p} \approx p_*$, the value segment is already arbitrarily close to flat, so it takes a vanishingly small stopping probability to turn the value segment completely flat. The same situation and intuition are repeated as we move backward in time: for a positive range of states, labeled Region 3, the DM repeatedly hedges between experimentation and Poisson stopping at a positive rate $\nu(\bar{p})$.

Intuitively, the repeated hedging by the DM can be seen as a resolution of conflicts between current and future selves. On the one hand, the DM is highly uncertain about the state and thus finds it optimal to acquire information. In particular, the DM worries about action ℓ being the wrong choice (state R) and would thus like to gather more information before taking it. On the other hand, the DM is aware that the journey of experimentation will take on its own life, with the future selves doing too much of it from the current self's perspective. Prolonged experimentation is particularly detrimental in state L, where the costly search for evidence is futile. The randomized stopping hedges the DM against both events: at each state in the mixing region, the DM's continuation value is independent of the underlying state. This resolves the conflict and restores the delicate balance between the multiple selves.

While at first glance it may seem odd that the randomized stopping arises in the middle region around p_*, this is no coincidence. As noted earlier, it is precisely where preference reversal and dynamic inconsistency arise. As will be seen, this seemingly peculiar feature of the equilibrium proves quite robust across different specifications of the learning environment and technologies.

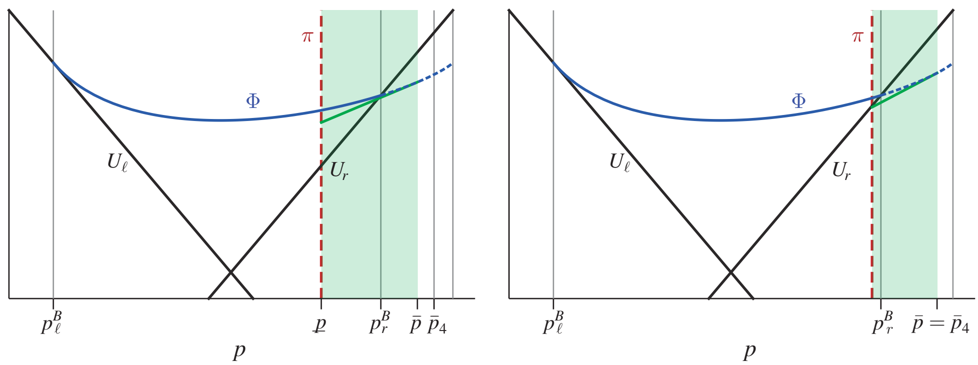

Figure 6. The Value Segment and Worst-Case Belief for $\bar{p} \in (\bar{p}_3, \bar{p}_4)$ in the Case of Δ Small

Region 4: $\bar{p} \in (\bar{p}_3, \bar{p}_4]$. As already mentioned, when ambiguity Δ is small, Region 3 is empty, and Regions 2 and 4 collapse to a single experimentation region. Its right boundary, $\bar{p}_4$, occurs at the point where, according to the worst-case belief $\underline{p}(\bar{p})$, action r yields the same expected payoff as experimentation, i.e., $U_r(\underline{p}(\bar{p}_4)) = V^\sigma(\underline{p}(\bar{p}_4), \bar{p}_4)$. Graphically, $\bar{p}_4$ can be seen as the point where the left endpoint of the value segment crosses the stopping payoff U_r (see right panel of Figure 6). It is important to notice that at $\bar{p} = \bar{p}_4$, the value of $\underline{p}(\bar{p})$ lies strictly to the left of the Bayesian stopping boundary p_r^B. This follows from the fact that the DM's future selves follow a rule that is optimal with respect to $\bar{p}$ but not with respect to $\underline{p}(\bar{p})$. From the perspective of $\underline{p}(\bar{p})$, the value of experimentation is thus strictly lower than the Bayesian value $\Phi(\underline{p}(\bar{p}))$. This means that the DM with worst-case belief $\underline{p}(\bar{p})$ is more inclined to stop than the Bayesian DM with the corresponding belief.

Next, consider the case in which Δ is sufficiently large so that Region 3 with mixed stopping is nonempty. The right boundary of the mixed stopping region, $\bar{p}_3$, is given by the state $\bar{p}$ at which the value segment separates from U_ℓ and experimentation becomes again optimal for all posteriors in $[\underline{p}(\bar{p}), \bar{p}]$ (see Figure 5). To the right of $\bar{p}_3$, we then have a second region of pure experimentation, Region 4. The value segment becomes upward sloping in this region, so $\underline{p}(\bar{p})$ becomes the worst-case belief. As with the case of small Δ, the right boundary of Region 4 is determined as the point where the left end of the value segment reaches the stopping payoff U_r. Notice, however, that the value segment in the experimentation region remains below the Bayesian value function. Compared to the case of small Δ, the possibility of randomization in future states reduces the value of experimentation further, which pushes the stopping boundary $\bar{p}_4$ toward the middle. Intuitively, the DM here is discouraged from experimentation and is thus more inclined to stop, not only because it leads to excessive experimentation in the distant future but also because it leads to inefficient stopping in the intermediate future.

We now complete the characterization by addressing several remaining cases and issues. The reader interested in the behavioral implications of the baseline characterization may skip to the next section.

Intermediate Experimentation Costs.—Consider next the case in which $c \in (\underline{c}, \bar{c})$. Theorem 1 shows that for each $c \in (\underline{c}, \bar{c})$, there exists a threshold $\Delta_c \in \mathbb{R}$ such that for $\Delta \leq \Delta_c$, the equilibrium continues to be described by Theorem 1(ii)(a). If instead $\Delta > \Delta_c$, the fourth region features randomization between actions ℓ and r rather than experimentation.

$$0 \underbrace{\qquad\qquad}_{\text{action } \ell} \bar{p}_1 \underbrace{\leftarrow\leftarrow\leftarrow\leftarrow}_{\text{pure exp.}} \bar{p}_2 \underbrace{\leftarrow\leftarrow\leftarrow\leftarrow\leftarrow}_{\text{mixed stopping } (\ell)} \bar{p}_3 \underbrace{\qquad\qquad\qquad}_{\text{mixing between } \ell \text{ and } r} \bar{p}_4 \underbrace{\qquad\qquad}_{\text{action } r} 1.$$

The equilibrium behavior for the first three regions is the same as in the case with low experimentation costs. In the third region, however, where the DM randomizes between experimentation and action ℓ, the value segment may drop as low as the point at which the two stopping payoff lines, U_ℓ and U_r, cross. The state at which this occurs constitutes our new boundary $\bar{p}_3$. The DM's value at this point is equal to $\min_p U(p) = U(\hat{p})$, the minimal expected payoff that the DM can guarantee to herself by randomizing between actions r and ℓ with probabilities $\hat{\rho}$ and $1 - \hat{\rho}$, respectively. At states to the right of this point, randomizing between immediate actions with $\hat{\rho}$ strictly dominates experimentation with randomized stopping, giving rise to the fourth region $[\bar{p}_3, \bar{p}_4)$. Its right boundary is the state $\bar{p}$ that satisfies $\underline{p}(\bar{p}) = \hat{p}$. Further to the right, where $\underline{p}(\bar{p}) > \hat{p}$, the DM optimally takes action r. The solution for intermediate costs $c \in (\underline{c}, \bar{c})$ and large ambiguity $\Delta > \Delta_c$ can thus be viewed as a combination of the previous two solutions.

Case 2.—We now turn to Case 2, which is characterized by $\Phi'(p_*) < 0$. The formal characterization of the intrapersonal equilibrium for this case is relegated to online Appendix B.8. Compared with the case $\Phi'(p_*) = 0$, there is only one significant change. At the right boundary of the first experimentation region $(\bar{p}_1, \bar{p}_2)$, the stopping distribution has an atom: the DM takes action r with positive probability $m(\bar{p}_2)$ and experiments with the complementary probability.

$$0 \underbrace{\qquad\qquad}_{\text{action } \ell} \bar{p}_1 \underbrace{\leftarrow\leftarrow\leftarrow\leftarrow}_{\text{pure exp.}} \underbrace{\bar{p}_2}_{m>0} \underbrace{\leftarrow\leftarrow\leftarrow\leftarrow}_{\text{mixed stopping } (\ell)} \bar{p}_3 \underbrace{\leftarrow\leftarrow\leftarrow\leftarrow\leftarrow}_{\text{pure exp. or mixing } \ell/r} \bar{p}_4 \underbrace{\qquad\qquad}_{\text{action } r} 1.$$

Moving backward in time through the experimentation region $(\bar{p}_1, \bar{p}_2)$, the value segment remains downward sloping as the state $\bar{p}$ approaches $\bar{p}_2 = p_*$. The unique worst-case belief in the limit as $\bar{p} \nearrow p_*$ is thus $\bar{p}$. For states to the right of p_*, the Bayesian optimal strategy under $\bar{p}$ is to take action r rather than experimenting. Taking action r in those states does, however, not constitute an intrapersonal equilibrium, as the worst-case belief under this strategy would be $\underline{p}(\bar{p})$ rather than $\bar{p}$. The DM must therefore experiment with positive probability in states to the right of p_*. To make experimentation optimal, the value segment must be weakly upward sloping for states belonging to an interval $(p_*, p_* + \varepsilon)$ so that nature is willing to choose a belief weakly below p_*. This property requires the DM to take action r with strictly positive probability at state $\bar{p} = p_*$. Graphically, the DM's randomization flips the left end of the value segment downward, so that it is flat at $\bar{p} = p_*$, as illustrated in Figure 7.[15]

[15]The knife-edge case $\bar{p}_0 = p_*$ (in Case 2) is the only instance where the requirement that the DM's strategy constitutes an equilibrium regardless of the initial state actually restricts the set of equilibria. In this instance, requiring equilibrium conditions *only* for subsequent states $\bar{p} < \bar{p}_0$ would leave some flexibility in picking the

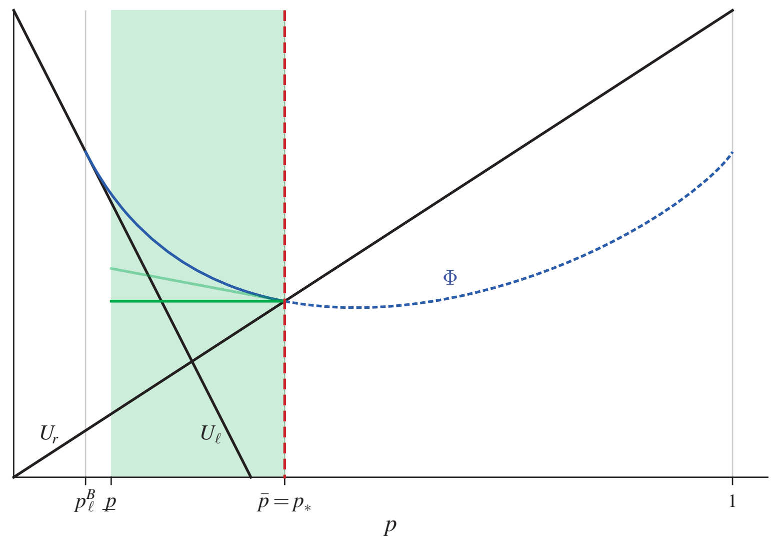

Figure 7. Case $\Phi'(p_*) < 0$: The Value Segment for $\bar{p} \nearrow p_*$ and $\bar{p} = p_*$

Uniqueness.—In online Appendix B.9, we prove that the intrapersonal equilibrium is unique when either $c < \underline{c}$ or $c \geq \bar{c}$. When costs are intermediate, however, there is a class of intrapersonal equilibria. These differ from the one described in Theorem 1 only in the position of their stopping boundary on the right. More specifically, recalling that $\hat{p}$ is the belief at which the DM is indifferent between actions ℓ and r, fix any q such that $\underline{p}(q) < \hat{p} \leq q$. We then construct a strategy profile $(\tilde{\sigma}, \tilde{\pi})$ indexed by $q \in [\hat{p}, \underline{p}^{-1}(\hat{p}))$ as follows:

$$\tilde{\sigma}(\bar{p}) = (\tilde{m}(\bar{p}), \tilde{\nu}(\bar{p}), \tilde{\rho}(\bar{p})) = \begin{cases} \sigma(\bar{p}); \\ (1, 0, \hat{\rho}); \\ (1, 0, 1); \end{cases} \quad \tilde{\pi}(\bar{p}) = \begin{cases} \pi(\bar{p}), & \text{if } \bar{p} \leq q; \\ \hat{p}, & \text{if } q < \bar{p} < \underline{p}^{-1}(\hat{p}); \\ \underline{p}(\bar{p}), & \text{if } \underline{p}^{-1}(\hat{p}) \leq \bar{p}; \end{cases}$$

where (σ, π) is the intrapersonal equilibrium defined in Theorem 1. In words, the DM follows the same strategy as in Theorem 1 for states $\bar{p} \leq q$ but stops immediately in states $\bar{p} > q$, followed by the mixed action $\hat{\rho}$ if $\hat{p} \in (\underline{p}(\bar{p}), \bar{p})$ and action r if $\hat{p} \leq \underline{p}(\bar{p})$. Since $c < \bar{c}$, this new strategy yields a strictly lower payoff for the DM in the states just to the right of q. In fact, the value function jumps down at the "switching" state $\bar{p} = q$ to the point where the two stopping payoffs cross, so the DM is worse off from this strategy relative to the one in Theorem 1. Nevertheless, the strategy profile—there is a continuum of such indexed by $q \in [\hat{p}, \underline{p}^{-1}(\hat{p}))$—satisfies all requirements of our solution concept, including the HJB conditions. The intuition is that, even though the HJB conditions capture control over vanishingly proximate future actions, these conditions are not strong enough. Given $c \geq \underline{c}$, there always

randomization between experimentation and action r at state $\bar{p}_0 = p_*$. Indeed, as long as the value segment remains (weakly) downward sloping, all equilibrium conditions would be satisfied for the states to the left of $\bar{p}_0 = p_*$. It is the equilibrium conditions for states $\bar{p} > p_*$ that require the value segment to become flat at state p_*.

exists some $\epsilon < \bar{p} - q$ such that deviating only until the state falls to $\bar{p} - \epsilon$ will not generate a strictly higher payoff.

Nevertheless, we view these equilibria as implausible. First, we see that $\tilde{\sigma}$, regardless of q, is Pareto worse than σ for all selves of the DM evaluated at the worst-case beliefs. Hence, $\tilde{\sigma}$ will be eliminated if the multiple selves of the DM can coordinate on the selection of equilibria, as is often invoked in the name of "consistent planning"; see Siniscalchi (2011).[16] Second, we could strengthen the notion of vanishing commitment to eliminate the equilibria. Namely, instead of requiring the equilibrium strategy profile to be a limit of the ϵ-commitment solution *pointwise* for each $\bar{p}$, one could require the limit convergence to be *uniform* across all $\bar{p}$'s. The profile $(\tilde{\sigma}, \tilde{\pi})$ cannot survive this stronger test since for any $\epsilon > 0$, there will always exist a state $\bar{p} \in (q, q + \epsilon)$, such that the ϵ-commitment solution for the DM at that state will entail a deviation to σ to enjoy a discontinuous payoff jump.[17]

Knightian Uncertainty.—The extreme case in which $\underline{p} = 0$ and $\bar{p} = 1$, known as Knightian uncertainty, constitutes an important theoretical benchmark. Naturally, the equilibrium characterization for this case is obtained as the limit of the solution we characterized above for $(\underline{p}_0, \bar{p}_0) \to (0,1)$.

PROPOSITION 1: *Assume $\mathcal{P}_0 = [0,1]$. If $c \geq \underline{c}$, the DM mixes between immediate actions ℓ and r with probability $\hat{\rho}$. Otherwise, the DM mixes between experimentation and stopping followed by ℓ at a stationary rate*

$$\tilde{\nu} = \frac{\lambda}{\delta_\ell}\left(u_r^R - \tilde{u}\right), \quad \textit{where} \quad \tilde{u} = \frac{u_r^R + u_\ell^L}{2} - \sqrt{\left(\frac{u_r^R - u_\ell^L}{2}\right)^2 + \frac{c}{\lambda}\delta_\ell}.$$

This solution is the limit of the equilibrium strategy characterized in Theorem 1 for the case $0 < \underline{p}_0 < \bar{p}_0 < 1$ as $(\underline{p}_0, \bar{p}_0) \to (0,1)$.

As $(\underline{p}_0, \bar{p}_0) \to (0,1)$, Region 3 featuring randomized stopping takes over the entire region to the right of p_*. Starting from $\bar{p}_0$ close to one, the DM will then spend an arbitrarily long time in Region 3. In the extreme case, when $[\underline{p}_0, \bar{p}_0] = [0,1]$, the DM's set of priors remains unchanged in the absence of breakthrough news. Hence, under any Markovian strategy, the DM either stops immediately or stops at a constant rate. The latter is optimal if and only if $c \leq \underline{c}$.

We can interpret the Knightian solution as a form of hedging by the DM to ensure her payoff equalizes between the states. Indeed, it is easy to verify that the stationary stopping rate $\tilde{\nu}$ satisfies the following equality:

$$u_\ell^L - \frac{c}{\tilde{\nu}} = \frac{\lambda}{\tilde{\nu} + \lambda} u_r^R + \frac{\tilde{\nu}}{\tilde{\nu} + \lambda} u_\ell^R - \frac{c}{\tilde{\nu} + \lambda}. \tag{17}$$

[16] More precisely, Siniscalchi (2011) requires the current self to break a tie in favor of her earlier self. See Ebert and Strack (2018) for a related approach.

[17] One could adopt this stronger requirement as a solution concept. We do not take this approach for two reasons. First, the uniform convergence requirement is not easy to check since it requires one to verify the equi-continuity property of the value function. Second, the requirement jeopardizes the existence in the more general Poisson model we discuss in online Appendix C. The unique intrapersonal equilibrium presented in Theorem 2 involves a discontinuous value function when the cost is intermediate, which does not survive the stronger refinement.

The LHS shows the DM's payoff from stopping at a constant rate $\tilde{\nu}$ when the state is L. In state L the DM never observes a breakthrough and thus always ends up taking the correct action. The downside is a relatively high expected cost of experimentation equal to $c/\tilde{\nu}$. The RHS shows the DM's payoff when the state is R. Now, the DM receives R-evidence before she stops experimenting, with positive probability $\lambda/(\tilde{\nu} + \lambda)$. With the complementary probability, however, the DM takes the wrong action and receives the lower payoff u_ℓ^R. The payoff gross of experimentation costs is then lower in state R than in state L, which is compensated by the DM experimenting for shorter periods (in expectation) and therefore incurring a lower expected cost of experimentation equal to $c/(\tilde{\nu} + \lambda)$.

The Role of Randomization.—Whether randomization helps to hedge against ambiguity depends on the DM's internal view on how the uncertainty unfolds (see Saito 2015 and Ke and Zhang 2020). We adopt the assumption that the DM views the game against nature as one with simultaneous moves. Hence, in the DM's mind, nature cannot condition her choice on the outcome of the DM's randomization. If the DM takes on an even more pessimistic view, believing that nature can adjust her choice *after* the DM's randomization outcome, any realization of a mixed strategy will be evaluated according to the worst-case belief for this realization, so the DM has no incentives to randomize.

Modifying the setting in this way clearly changes the characterization of our solution for the case when ambiguity is large. It does not, however, erase the DM's desire to stop preemptively in anticipation of a long experimentation phase when ambiguity is large. As we will show, stopping occurs again at intermediate sets of beliefs where the worst-case belief is about to switch. But instead of an extended period of randomized stopping for an interval of states, now stopping occurs at an isolated state $\bar{p}^*$. Anticipating this, earlier selves of the DM are happy to experiment until that point. Even though the game ends once the state reaches $\bar{p}^*$, if the DM would find herself at state $\bar{p} < \bar{p}^*$, she would again be experimenting. In this sense, the nonmonotonic pattern of stopping persists even without randomization. To keep the proof short, we restrict attention to the symmetric case where $u_r^R = u_\ell^L =: \delta$ and $u_\ell^R = u_r^L = 0$.

PROPOSITION 2: *Suppose the DM is restricted to pure strategies. Assume $\Phi'(p_*) = 0$, $c < \bar{c}$, and payoffs are symmetric. If Δ is sufficiently large, then there is a state $\bar{p}^* \in (0,1)$ such that the DM takes action ℓ at $\bar{p}^*$ and experiments on a left and right neighborhood of $\bar{p}^*$.*

To understand this result, let us meet the DM on the journey backward in time, at the exact moment when the worst-case belief is about to switch, which is captured by the right panel of Figure 4.[18] Moving further back in time from that moment, the value segment associated with experimentation becomes upward sloping, so the DM, as before, finds herself in a dilemma where further experimentation is met with the adversarial belief $\underline{p}(\bar{p})$, whereas stopping with ℓ is met with $\bar{p}$. With nature now

[18] Note the DM's equilibrium behavior in Regions 1 and 2 is unchanged.

wielding the second-mover advantage, the DM can no longer protect herself from these adversarial beliefs via randomization. Instead, the DM must choose the lesser of two bad choices. As long as $\bar{p} > p_*$ is relatively close to p_*, the value segment is sufficiently flat so that the value of experimentation at $\underline{p}(\bar{p})$ is higher than the stopping payoff U_ℓ at belief $\bar{p}$. Experimentation thus remains the better option for an interval of time before p_* is reached. However, going further back in time, and assuming that ambiguity is sufficiently large, there comes a state $\bar{p}^*$ at which the value of experimentation at its worst belief $\underline{p}(\bar{p}^*)$ is as low as $U_\ell(\bar{p}^*)$. At that state, stopping becomes optimal for the DM. As noted above, given that the DM stops at $\bar{p}^*$, earlier selves are willing to experiment until $\bar{p}^*$ is reached. The equilibrium strategy thus prescribes experimentation to the left and right of $\bar{p}^*$.

III. Behavioral Implications of Ambiguity

In this section, we perform comparative statics to further explore the behavioral implications of ambiguity.

Prolonged Learning and Indecision.—We first investigate the DM's incentive to prolong her experimentation late in the game and how it varies with ambiguity. For the latter purpose, we denote by θ the *true* probability of R and consider the case where the DM's set of priors $\mathcal{P}$ is centered around θ. Specifically, for each $\theta \in (0,1)$ and each $\Delta \geq 0$, we define

$$\mathcal{P}_\Delta(\theta) := \left\{p \in [0,1] : \left|\ln\left(\frac{p}{1-p}\right) - \ln\left(\frac{\theta}{1-\theta}\right)\right| \leq \frac{\Delta}{2}\right\}$$

as the set of priors with midpoint θ in terms of the log-likelihood ratio and ambiguity level Δ. We then denote by $T_\Delta(\theta)$ the expected length of experimentation when the DM's set of priors is $\mathcal{P}_\Delta(\theta)$ and the true probability of state R is θ. We have the following result.

PROPOSITION 3 (Length of Experimentation): *Suppose $c < \bar{c}$ so that experimentation occurs under some θ. Let $0 \leq \Delta < \Delta'$, and define $\bar{\theta} := \sup\{\theta : T_\Delta(\theta) > 0\}$. There exists some $\hat{\theta} \in (0,\bar{\theta}]$ such that $T_{\Delta'}(\theta) \geq T_\Delta(\theta)$ if $\theta < \hat{\theta}$ and $T_{\Delta'}(\theta) \leq T_\Delta(\theta)$ if $\theta \in [\hat{\theta},\bar{\theta})$. Further, $dT_{\Delta'}/d\theta \leq dT_\Delta(\theta)$ for all $\theta < \hat{\theta}$.*

In words, an increased ambiguity leads the DM to prolong her experimentation when the likelihood θ of state R is sufficiently small. For a benchmark, consider the Bayesian DM with accurate prior $p_0 = \theta$ and no ambiguity ($\Delta = 0$). The length of experimentation for the Bayesian DM is concave and strictly increasing in θ for θ not too large,[19] as illustrated in Figure 8. There are two competing forces. On the one hand, as θ falls, experimentation is less likely to produce conclusive R-evidence; this tends to lengthen the experimentation. On the other hand, the DM adjusts her learning strategy to shorten its duration as her (accurate) prior θ falls. For θ

[19] See Proposition 1 of Che and Mierendorff (2019).

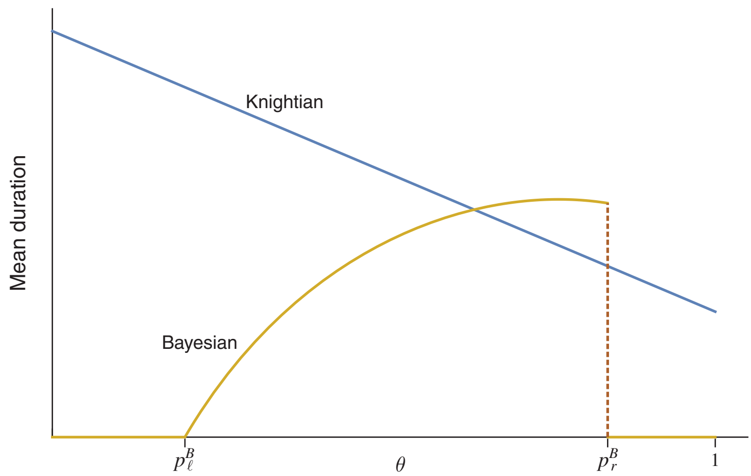

Figure 8. Mean Sampling Time for the Bayesian Case and the Case of Knightian Uncertainty

sufficiently small, the latter effect dominates the former effect, so the overall learning time decreases as R becomes less likely.

As stated in Proposition 3, ambiguity makes the DM's strategy less responsive to the change in θ. The reduced responsiveness to θ corresponds to the behavioral trait of "indecisiveness." In the extreme case of $\Delta \to \infty$ (i.e., Knightian ambiguity), the learning strategy becomes completely unresponsive to the true likelihood θ, so that, as θ falls, her learning time actually increases (see Figure 8). The reason is that even after an arbitrarily long duration of unsuccessful experimentation, the DM's extreme ambiguity prevents her from dismissing R as sufficiently unlikely and stopping experimentation.

Premature Decisiveness.—While the Bayesian optimal stopping time is deterministic conditional on not receiving R-evidence, a sufficiently large ambiguity leads to randomized stopping at earlier points in time. Therefore, the ambiguity-averse DM exhibits early "decisiveness" not shown by her Bayesian counterpart. At the same time, an increased ambiguity also leads to prolonged experimentation, as seen in Proposition 3, which increases the likelihood of delayed stopping. The combined effect is the increased dispersion of the stopping times. To formalize this result, consider two prior sets $\mathcal{P}$ and $\mathcal{Q}$ with $\mathcal{P} \subset \mathcal{Q}$. Let $F_{\mathcal{P}} : [0, +\infty) \to [0,1]$ denote the cumulative distribution function over stopping times when the initial set of priors is $\mathcal{P}$. Hence, given prior set $\mathcal{P}$, $F_{\mathcal{P}}(t)$ is the probability that the DM stops before time t.

PROPOSITION 4 (Dispersion of Stopping Times): *Assume $c \leq \underline{c}$ and $\Phi'(p_*) = 0$. Let $\mathcal{P}$ and $\mathcal{Q}$ be such that $\mathcal{P} \subseteq \mathcal{Q}$. There exists a time $\hat{t}$ such that $F_{\mathcal{Q}}(t) \geq F_{\mathcal{P}}(t)$ for all $t < \hat{t}$ and $F_{\mathcal{Q}}(t) \leq F_{\mathcal{P}}(t)$ for all $t > \hat{t}$.*

Proposition 4 shows that the stopping-time distribution under the two nested sets $\mathcal{P}$ and $\mathcal{Q}$ exhibits a single crossing property. This property indicates that the distribution over stopping times under the larger set $\mathcal{Q}$ is more dispersed than the one

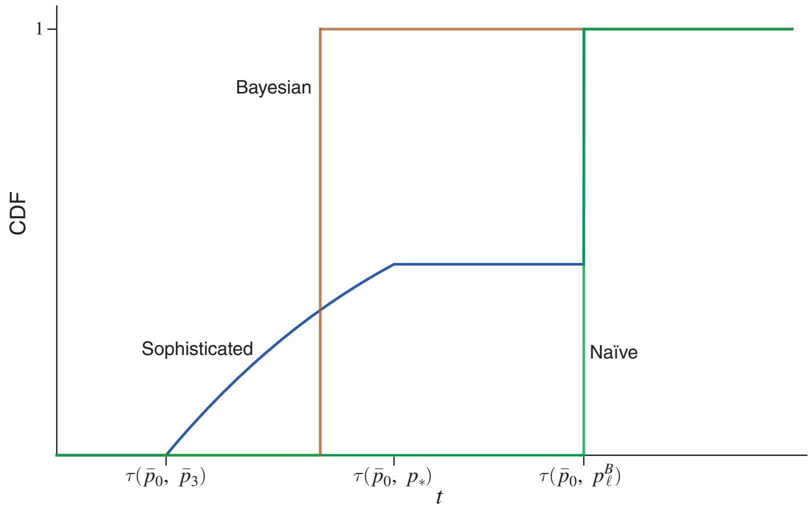

Figure 9. Cumulative Distribution of Stopping Times (Conditional on $\omega = L$) for the Bayesian DM and the Ambiguity-Averse DM When Sophisticated (Blue) and Naïve (Green)

Note: In the graph, $\tau(p,q)$ denotes the time it takes for belief p to be updated to q in the absence of R-evidence.

under $\mathcal{P}$. Indeed, whenever the mean stopping time under $\mathcal{P}$ is weakly greater than under $\mathcal{Q}$, then $F_{\mathcal{P}}$ second-order stochastically dominates $F_{\mathcal{Q}}$ (when the means are equal, $F_{\mathcal{Q}}$ constitutes a mean-preserving spread of $F_{\mathcal{P}}$). The single crossing property implies that the range of potential stopping points spreads as ambiguity becomes larger: an expansion of the initial set of priors leads to an earlier first potential stopping point and a later last potential stopping point. The property is illustrated in Figure 9, where we plot the cumulative probability of stopping conditional on state L for the Bayesian DM (orange graph) and the ambiguity-averse DM (blue graph).

Can the Two Traits Coexist?—According to our theory, ambiguity aversion leaves potentially testable behavioral markers—prolonged indecision and impulsive decisiveness. While seemingly contradictory, there is a coherent logic that ties these two traits together. For an illustration, consider again the theorist seeking to prove a new theorem with unknown validity. According to our theory, ambiguity—with regard to the unknown validity of the theorem—makes it difficult for the theorist to dismiss the validity of the theorem even after many unsuccessful attempts to prove it. Hence, ambiguity might lead the DM to a costly process of continued trial that could be seen by the outsider as a "wild goose chase." Recognizing that such an obsessive pursuit is in the offing—perhaps also reminded by how such endeavors ended in the past—the DM may simply refrain from getting into the situation by stopping early; the theorist may prematurely give up after making a few unsuccessful attempts at proving the theorem.[20]

[20] Barkley-Levenson and Fox (2016, p. 5) find indecision and impulsivity to be positively correlated among the respondents of their behavioral survey and reach an interpretation uncannily close to our theory: "Our findings suggest … indecisiveness and impulsivity can be viewed as two sides of the same coin. Both indecisiveness and impulsivity are maladaptive behavioral responses to difficulty engaging with a decision. We surmise that these responses arise from a common desire to avoid negative affect that some individuals experience when making

The Role of Sophistication.—While the feature of prolonged learning as a response to ambiguity does not hinge on the DM's sophistication, premature decisiveness is driven by the DM's ability to correctly forecast her future behavior and respond accordingly. Indeed, a fully naïve DM—one who acts according to the best plan given by the current worst-case scenario but fails to carry out the plans of earlier selves—only exhibits the first behavioral trait and will thus always experiment longer than the corresponding Bayesian DM. The sophisticated DM, on the other hand, will engage in preemptive stopping when ambiguity is sufficiently large, so she will tend to stop earlier than her naïve counterpart. Note, however, that both DMs share the same final stopping points, so with positive probability both stop at the same time. We illustrate this feature in Figure 9: the CDF for the naïve DM first-order stochastically dominates that of the sophisticated DM.

It is also interesting to contrast the sophisticated and naïve DM with a DM who has full commitment power. As we saw in Section I, the commitment solution is given by the Bayesian optimal stopping rule for the prior in $[\underline{p}_0, \bar{p}_0]$ that minimizes the Bayesian value function Φ^*. Assuming $\bar{p}_0 > p_*$, the graph illustrating the commitment solution is thus qualitatively similar to that of the Bayesian DM in Figure 9. Specifically, a DM with full commitment will stop before the naïve DM does. The comparison with a sophisticated DM is ambiguous, except that the latter's stopping time is more dispersed in the sense established in Proposition 2.

IV. Incremental Learning

So far, we have focused on Poisson learning, where news is one-sided and conclusive. Another canonical case is the incremental learning model in which the signal follows a diffusion process. In this section, we modify the baseline model and consider this alternative information technology. For simplicity, we restrict attention to the case where payoffs are symmetric across states: $u_r^R = u_\ell^L, u_r^L = u_\ell^R$.

Suppose the DM receives a signal $X \in \mathbb{R}$ about the state $\omega \in \{L, R\}$ at a flow cost of c per unit time, which follows the process

$$dX_t = \mu_\omega dt + \sigma dB_t,$$

where $\mu_R > \mu_L$. As usual, it is more convenient to convert X into a belief, particularly into a log-likelihood ratio Z, where

$$dZ_t = \frac{\psi}{\sigma} dX_t - \frac{\psi}{\sigma}\left(\frac{\mu_R + \mu_L}{2}\right) dt = \pm \frac{\psi^2}{2} dt + \psi dB_t,$$

and $\psi := (\mu_R - \mu_L)/\sigma$ is the signal-to-noise ratio (see Daley and Green 2012, 2020, for example).

As before, the Bayesian optimal behavior is characterized by two stopping boundaries (in terms of log-likelihood ratio), z_ℓ and z_r, such that the DM stops and chooses ℓ if $z \leq z_\ell$, stops and chooses r if $z \geq z_r$, and experiments if $z \in (z_\ell, z_r)$.

choices. … they may behave in an impulsive manner, hastily choosing in order to avoid unpleasant deliberation over opportunity costs."

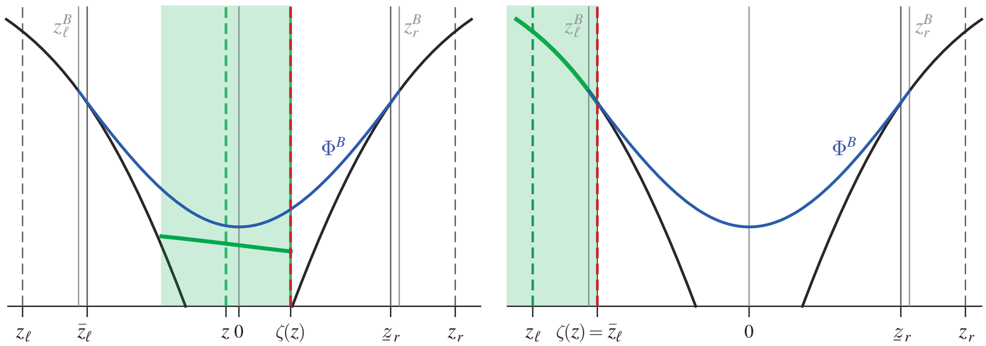

Figure 10. Agent's Value when Experimenting (Left) and When Stopping (Right), with Worst-Case Belief $\zeta(z)$ and Boundaries $\bar{z}_\ell := z_\ell + \Delta/2$ and $\underline{z}_r := z_r - \Delta/2$

Note: The Bayesian value function is $\Phi^B(z) := \phi(z \mid z_\ell^B, z_r^B)$.

Let $\phi(z \mid z_\ell, z_r)$ be the value of this strategy when the DM's prior is z, where $z = 0$ corresponds to the belief of $p = 1/2$. The Bayesian optimal stopping boundaries are denoted by (z_ℓ^B, z_r^B), where $z_\ell^B = -z_r^B < 0$ due to the symmetry of payoffs. As before, the stopping boundaries are graphically depicted by tangent points of the Bayesian value function at the stopping payoffs (which are nonlinear due to the change of variable); see Figure 10.[21]

Consider now an ambiguity-averse DM whose prior set is represented by an interval $\mathcal{Y}(z) := [z - \Delta/2, z + \Delta/2]$. In contrast to before, let us index the "state" by the midpoint z and the deviation from the midpoint in terms of the log-likelihood ratio by $\gamma \in [-\Delta/2, \Delta/2]$. Upon suitable modification, the intrapersonal equilibrium is characterized much like in the baseline model.

Small Δ.—If ambiguity is sufficiently small, there are again two stopping boundaries $z_\ell < 0 < z_r$ (where $z_r = -z_\ell$) such that the DM experiments if and only if the state z belongs to the interval (z_ℓ, z_r). Suppose the DM follows this strategy. Then, at state z, the DM with belief $(z + \gamma) \in \mathcal{Y}(z)$ perceives an expected payoff of

$$V(z + \gamma, z) = \phi(z + \gamma \mid z_\ell + \gamma, z_r + \gamma).$$

Close to the stopping boundary, the DM's worst-case belief is given by the "innermost" belief of the set, just as in the Poisson model. The two stopping boundaries z_ℓ and z_r are thus determined by the smooth-pasting conditions with respect to these beliefs:

$$\phi'\left(z_\ell + \frac{\Delta}{2} \,\middle|\, z_\ell + \frac{\Delta}{2}, z_r + \frac{\Delta}{2}\right) = U_\ell'\left(z_\ell + \frac{\Delta}{2}\right),$$
$$\phi'\left(z_r - \frac{\Delta}{2} \,\middle|\, z_\ell - \frac{\Delta}{2}, z_r - \frac{\Delta}{2}\right) = U_r'\left(z_r - \frac{\Delta}{2}\right),$$

[21] This value function and the Bayesian optimal stopping boundaries, z_ℓ^B and z_r^B, can be derived using standard methods.

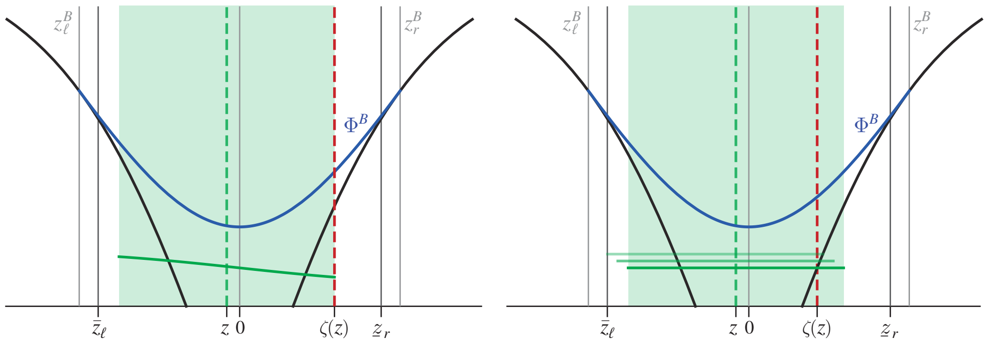

Figure 11. The Case of Large Δ: On the Left, Profitable Deviation if the DM Would Experiment; on the Right, Intrapersonal Equilibrium with Mixed Stopping (Followed by r)

where, with slight abuse of notation, $U_a, a = \ell, r$ denotes the DM's stopping payoff as a function of the log-likelihood ratio. Stopping for action ℓ occurs at state z_ℓ where the DM is indifferent between stopping and experimenting *at her rightmost belief* $z_\ell + \Delta/2$, while stopping for action r occurs at state z_r where the DM is indifferent between stopping and experimenting *at her leftmost belief* $z_r - \Delta/2$.

A notable difference relative to the Poisson model is that the stopping beliefs, $z_\ell + \Delta/2$ and $z_r - \Delta/2$, no longer coincide with the corresponding Bayesian stopping boundaries, z_ℓ^B and z_r^B. Instead, we have $z_\ell^B < z_\ell + \Delta/2 < 0 < z_r - \Delta/2 < z_r^B$, so the DM stops experimenting before stopping becomes Bayesian optimal for all beliefs in the set.[22] To see this, consider the case where the DM's rightmost belief is close to z_ℓ^B. If the DM continues experimentation, there is a small chance that the state drifts all the way to the other side of the right stopping boundary. The DM understands that if this happens, she will stop only when the leftmost belief reaches the stopping boundary. From the viewpoint of the rightmost belief, there is hence excessive experimentation close to the right stopping boundary, which reduces the value of experimenting. The same applies to the other side. The argument further implies that in the experimentation region, there is no belief in $[z - \Delta/2, z + \Delta/2]$ for which the equilibrium strategy is Bayesian optimal. This means that the value segment for the ambiguity-averse DM is strictly below, and thus never touches, the Bayesian value function, as illustrated in Figure 10.

Large Δ.—The above solution constitutes an equilibrium only when ambiguity is sufficiently small. Graphically, the value segment has to be sufficiently short so that its lower end does not fall below the stopping payoff. Otherwise, stopping becomes optimal under the worst-case belief, as we illustrate in Figure 11 (left panel). In such situations, the DM again randomizes between experimentation and stopping. Mixed stopping works here as follows: starting from $z = 0$, if z moves slightly to the left, the DM mixes between experimentation and stopping followed by action r; if instead z moves slightly to the right, the DM mixes between

[22] Epstein and Ji (2022) obtain a qualitatively similar result for the case of rectangular sets of priors.

experimentation and stopping followed by ℓ. This pattern is analogous to how randomized stopping occurs in the baseline model. Intuitively, mixing with action r when $z < 0$ (respectively, ℓ when $z > 0$) hedges the DM against the possibility that the state is in fact R (respectively, L) and, hence, that experimentation will take a very long time.

The DM's value and equilibrium stopping rate in the mixed stopping region of the drift-diffusion model can be derived via the same approach we developed for the model with Poisson learning. The saddle point HJB condition is now

$$c = \max_{\nu} \min_{y \in \mathcal{Y}(z)} \nu\big(U(y) - V(y,z)\big) + \frac{\psi^2}{2}\big[2p(y) - 1\big]V_y(y,z)$$
$$+ \frac{\psi^2}{2}\big[2p(y) - 1\big]V_z(y,z) + \frac{\psi^2}{2}\big[V_{yy}(y,z) + V_{yz}(y,z) + V_{zz}(y,z)\big],$$

where $p(y) := e^y/(1 + e^y)$ is the probability of R corresponding to the log-likelihood ratio y. Letting $\zeta = \zeta(z)$ denote nature's choice in state z, we can derive $\zeta(z)$ and the DM's stopping rate $\nu(z)$ by following the same steps as in Section IIB. Let us consider the case $z < 0$, where stopping is followed by r (the case $z > 0$ is analogous). Randomization between experimentation and r requires that the coefficient of ν must vanish:

$$U_r\big(\zeta(z)\big) = V\big(\zeta(z), z\big). \tag{18}$$

To satisfy this condition, nature must choose an interior $\zeta(z)$, which requires that the value segment is flat. This means that V_y, V_{yz}, V_{yy} all vanish and we can write $V(y,z) = \hat{V}(z)$ for all y. We then use equation (18) to pin down nature's choice:

$$\zeta(z) = p^{-1}\left(\frac{\hat{V}(z) - u_r^L}{u_r^R - u_r^L}\right). \tag{19}$$

For nature to choose an interior likelihood ratio, the coefficient of y in the saddle point condition must vanish, too:

$$\nu(z)\big(u_r^R - u_r^L\big) + \psi^2 V_z\big(\zeta(z), z\big) = 0. \tag{20}$$

Using again equation (18), this requirement ties down the DM's stopping rate $\nu(z)$. Finally, we need to determine the value function $\hat{V}(z)$. To this end, we substitute for $\zeta(z)$ and $\nu(z)$ to simplify the saddle point condition and obtain the analog of ($\widehat{\text{ODE}}$).

$$c = \psi^2\left(\frac{\hat{V}(z) - u_r^L}{u_r^R - u_r^L} - \frac{1}{2}\right)\hat{V}'(z) + \frac{\psi^2}{2}\hat{V}''(z).$$

The main complication in the drift-diffusion model is that the boundary condition for this ODE is no longer pinned down by the minimum of the Bayesian value function, as it was before, but has to be found as a fixed point.

Apart from this technical difference, the solution method and qualitative features of the solution extend to the drift-diffusion model. In particular, (i) the DM has incentives to engage in excessive experimentation close to the stopping boundaries; (ii) if ambiguity is large and the flow cost c is small, the DM will exert self-control and preempt herself from experimenting excessively by randomly stopping at intermediate sets of beliefs.

V. Concluding Remarks

This paper studies the canonical Wald problem under the assumption that the DM faces uncertainty about the underlying probabilities of the different states. We consider a DM who seeks to maximize her payoff guarantee and updates beliefs prior by prior. Since the DM's worst-case belief may change over time, the DM's preference (at the worst-case belief) may violate dynamic consistency. Rather than ruling out such violations, we seek to understand their behavioral implications on information acquisition. The main technical challenge is then to solve the DM's optimization problem in the presence of dynamic preference reversals. We develop a new method, based on a version of HJB conditions, and provide a microfoundation for the approach. The method proves highly tractable and is applicable to dynamic optimization problems beyond information acquisition, such as real options theory, dynamic portfolio choice, or search models.

We show that ambiguity gives rise to two behavioral traits—prolonged indecision and premature decisiveness. The inability by our DM to stop searching for R-evidence after many unsuccessful attempts is reminiscent of the *sunk cost fallacy*, i.e., the tendency not to give up on projects or activities in which one has invested. In our case, the DM struggles to dismiss the possibility of the state being R even after many failed bids to find evidence for it. The DM in that situation is guided by the rightmost belief—namely, the belief that not only rationalizes the incurred experimentation costs so far but also calls for further continuation of learning. Ambiguity aversion could thus be viewed as an alternative mechanism behind the sunk cost fallacy, complementing other existing behavioral explanations.

Our model can be extended to allow for a choice in the type of information the DM acquires. In online Appendix C we consider the general Poisson model introduced by Che and Mierendorff (2019), where the DM has access to two news sources, one emitting R-evidence (as in the baseline model), the other emitting L-evidence. Conditional on acquiring information, the DM must then allocate her attention between the two news sources. For instance, a theorist may try either to "prove" a theorem (R-evidence), to find a "counterexample" disproving it (L-evidence), or to divide effort between the two endeavors. We show that excessive learning and randomized stopping continue to arise in this setting with sufficiently large ambiguity and sufficiently small information costs. In addition, the DM may "split" her attention between the two news sources in order to hedge, even when a Bayesian DM would not do so under any prior. Whenever the DM employs the split-attention strategy, she continues learning until she fully resolves the uncertainty. This behavior thus provides another pattern of excessive learning, when it is seen from the perspective of a Bayesian observer with comparable beliefs.

An open question is how our solution changes under different updating rules. Prior-by-prior updating entails that the DM updates on the state but not on the probability model. On the other extreme of the spectrum is maximum-likelihood updating (Gilboa and Schmeidler 1993), according to which the DM only retains those priors that explain the data best. In our setting, this would not be very interesting, as the DM, in the absence of a breakthrough, would discard all priors but the leftmost after the first instant of time. There are, however, less extreme versions that combine features of both updating rules (e.g., Cheng 2021). It will be interesting to explore the implications of such rules on the incentives to acquire information in dynamic settings.

We focus on the case where the DM is fully sophisticated and thus correctly forecasts the behavior of her future selves. An open question is how the results extend to the case where the DM is partially naïve. For instance, the DM may be paying attention only at certain points in time. This could be modeled by having a Poisson clock run in parallel, which then determines the times at which the DM contemplates the actions of her future selves and optimizes accordingly (otherwise, she follows the naïvely optimal plan). A higher Poisson rate would thus capture a more sophisticated DM. Given such a model, one could ask how the DM's welfare (say, with respect to initial preferences) would depend on the level of sophistication. We leave these interesting questions for future research.

Mathematical Appendix

A1. *Markov Strategies and Their Values*

For any fixed prior set $\mathcal{P}_0 = [\underline{p}_0, \bar{p}_0]$, we first express the DM's strategy as specifying her action after each time t of experimenting without receiving any news. (Recall that the latter are the only nontrivial subgames we focus on since the DM chooses r immediately following the arrival of R-evidence.)

Specifically, a *time-indexed strategy* is a function $\tilde{\sigma} : [0, \infty) \to [0, \infty) \times [0,1]^2$, $\tilde{\sigma}_t = (\tilde{\nu}_t, \tilde{m}_t, \tilde{\rho}_t)$, where $\tilde{\nu}_t$ is the stopping rate and $\tilde{m}_t$ the instantaneous stopping probability used at time t if no R-signal has been received up until time t. $\tilde{\rho}_t$ is the probability of action r conditional on stopping at time t without receiving an R-signal.[23] To ensure that a strategy admits a well-defined stopping time, we require that (i) any connected set of t's in which $\tilde{m}_t = 1$ contains its infimum and (ii) the set $\{t \geq 0 \,|\, \tilde{m}_t \in (0,1)\}$ is countable. Time-indexed strategies satisfying these restrictions are called *admissible*, and $\tilde{\Sigma}$ denotes the set of all admissible time-indexed strategies. Fix any $\tilde{\sigma} \in \tilde{\Sigma}$. Then, starting at any time s, the probability that the strategy $\tilde{\sigma}$ prescribes the DM to stop by time $t > s$ is written as

$$F_s(t - s) := 1 - e^{-\int_s^t \tilde{\nu}_{s'} ds'} \prod_{s' \in [s,t]} (1 - \tilde{m}_{s'}).$$

[23] Online Appendix B.3 shows that the current formulation of a strategy is induced by a well-behaved stopping-time distribution that the DM may choose.

Using this function, we can write the expected payoff, or the *value*, of strategy $\tilde{\sigma}$ given belief p at time t as follows:

$$\text{(A1)} \qquad \tilde{V}^{\tilde{\sigma}}(p,t) = p\left\{ \int_t^{\infty} [1 - F_t(s)] e^{-\lambda s} [\tilde{\nu}_s U^R(\tilde{\rho}_s) + \lambda u_r^R - c] ds + \sum_{s \in [t,\infty)} [1 - F_t(s)] \tilde{m}_s e^{-\lambda s} U^R(\tilde{\rho}_s) \right\} + (1-p)\left\{ \int_t^{\infty} [1 - F_t(s)] e^{-\lambda s} [\tilde{\nu}_s U^L(\tilde{\rho}_s) - c] ds + \sum_{s \in [t,\infty)} [1 - F_t(s)] \tilde{m}_s U^L(\tilde{\rho}_s) \right\},$$

where $U^{\omega}(\rho) = \rho u_r^{\omega} + (1-\rho) u_{\ell}^{\omega}$ denotes the utility from the mixed action ρ if the state is ω.

The law of motion for the belief in the absence of stopping is exogenous and deterministic, and the belief always drifts downward. Hence, there is a one-to-one mapping between time t and the belief set $\mathcal{P}_t := [\underline{p}_t, \bar{p}_t]$ for any initial prior belief set $\mathcal{P}_0 := [\underline{p}_0, \bar{p}_0]$ in the absence of stopping, *independent of the DM's strategy*. That is, both $\underline{p}_t$ and $\bar{p}_t$ drift down according to the law of motion (1) given the initial values, $\underline{p}_0$ and $\bar{p}_0$. As in the text, we index the set $\mathcal{P}_t$ by the highest possible belief $\bar{p}_t$ and call it the *state*. Noting that given $\bar{p}$, $\underline{p}$ is uniquely pinned down, we denote $\underline{p}_t := \underline{p}(\bar{p}_t)$. Any $\bar{p} < \bar{p}_0$ can then be uniquely mapped to the time $t \geq 0$ it takes for the state to drift down to $\bar{p}$:

$$\text{(A2)} \qquad t = \tau(\bar{p}) := \frac{1}{\lambda} \ln\left(\frac{\bar{p}}{1-\bar{p}} \frac{1-\bar{p}_0}{\bar{p}_0} \right).$$

Consequently, any time-indexed strategy $\tilde{\sigma} = (\tilde{\sigma}_t)$ induces a strategy $\sigma = (\sigma(\bar{p})) := (\tilde{\sigma}_{\tau(\bar{p})})$, and conversely, any strategy induces a time-indexed strategy via $\tilde{\sigma}_t := \sigma(\tau^{-1}(t))$. Importantly, note that the Markov structure of the strategy—the dependence of the strategy only on the state—involves no restriction on the strategy space.[24] Clearly, the admissibility on $\tilde{\sigma}$ corresponds to the admissibility imposed on σ. The expected payoff from any $\sigma \in \Sigma$ can then be computed from (A1) via a change of variable using (A2):

$$\text{(A3)} \qquad V^{\sigma}(p,\bar{p}) := \tilde{V}^{\tilde{\sigma}_{\tau(\cdot)}}(p, \tau(\bar{p})).$$

A2. *Microfoundation of the Saddle Point HJB Equations*

We now justify our equilibrium concept or the use of saddle point HJB equations. The idea is to imagine that at each point in time, the DM can control, or commit to, her

[24] Recall that this one-to-one mapping is a result of our environment. Beyond our setting, it may be restrictive to focus on Markov strategies. That is, if the mapping $t \mapsto \bar{p}_t$ is not injective, then there can be strategies that depend on calendar time that cannot be encoded as a Markov strategy.

actions for a brief period of time, say, $\epsilon > 0$. She thus acts dynamically consistently for a brief period of time, while taking as given the strategies of her future selves beyond that time interval. We then require the strategy to be a limit of the ϵ-dynamically consistent strategies as $\epsilon \to 0$.

To make this formal, fix a candidate strategy σ^*. Suppose that, at any state $\bar{p}$ and any time t, the DM can commit to any strategy σ (possibly different from σ^*) during a time interval $[t, t+\epsilon)$, for some $\epsilon > 0$, and the strategy switches to σ^* thereafter. Let $(\sigma_\epsilon(\cdot), \pi_\epsilon(\cdot))$ be the saddle point strategy for such an ϵ horizon problem (followed by σ^*), or an ϵ-commitment solution. We say that a pseudo equilibrium (σ^*, π^*) is *dynamically credible* if for each $\bar{p}$ there is a sequence of ϵ-commitment solutions $(\sigma_\epsilon(\bar{p}), \pi_\epsilon(\bar{p}))$ that converges to (σ^*, π^*) as $\epsilon \to 0$. We further say that a pseudo equilibrium (σ^*, π^*) is *well behaved* if, for each $\bar{p} \in (0,1)$, $V^{\sigma^*}(\pi^*(\bar{p}), \bar{p})$, $V_p^{\sigma^*}(\pi^*(\bar{p}), \bar{p})$, and $V_p^{\sigma^*}(\pi^*(\bar{p}), \bar{p}_-)$ are well defined and continuous at $(\pi^*(\bar{p}), \bar{p})$. We now provide the justification based on dynamic consistency.

Write $W_\epsilon^\sigma(p, \bar{p})$ for the DM's value when the DM follows strategy σ for ϵ-period, followed by the candidate strategy σ^*. Then,

$$\begin{aligned} W_\epsilon^\sigma(p,\bar{p}) \;=\; & m(\bar{p})\,U_{\rho(\bar{p})}(p) + \big[1 - m(\bar{p})\big] \\ & \times \Bigg\{ -p\int_{\bar{p}^\epsilon}^{\bar{p}} e^{-\int_0^{\tau(\bar{p},\bar{p}')}[\lambda+\nu(\bar{p}^\tau)]d\tau}\Big[-c + \lambda u_r^R + \nu(\bar{p}')u^R_{\rho(\bar{p}')}\Big]\frac{1}{\eta(\bar{p}')}d\bar{p}' \\ & \quad - (1-p)\int_{\bar{p}^\epsilon}^{\bar{p}} e^{-\int_0^{\tau(\bar{p},\bar{p}')}\nu(p^\tau)d\tau}\Big[-c + \nu(\bar{p}')u^L_{\rho(\bar{p}')}\Big]\frac{1}{\eta(\bar{p}')}d\bar{p}' \\ & \quad + \Big[p e^{-\int_0^{\tau(\bar{p},\bar{p}^\epsilon)}[\lambda+\nu(p^\tau)]d\tau} + (1-p)e^{-\int_0^{\tau(\bar{p},\bar{p}^\epsilon)}\nu(p^\tau)d\tau}\Big] V^{\sigma^*}(p^\epsilon, \bar{p}^\epsilon) \Bigg\}, \end{aligned}$$

where for any $p \in [0,1]$, p^τ denotes the τ-ahead update of belief p and

$$\tau(p,p') \;:=\; \frac{1}{\lambda}\ln\left(\frac{p}{1-p}\,\frac{1-p'}{p'}\right)$$

is the time it takes for the Bayesian update of p to reach p' in the absence of R-evidence. (Note that the earlier definition of $\tau(\bar{p})$ in Appendix A1 suppresses the first argument since it was fixed to $\bar{p}_0$.) Let $(\sigma_\epsilon(\cdot), \pi_\epsilon(\cdot))$ denote an ϵ-commitment solution. Then, it must satisfy the following saddle point condition: for each $\bar{p}$,

$$\text{(A4)} \qquad \sigma_\epsilon(\bar{p}) \;\in\; \arg\max_{\sigma} W_\epsilon^\sigma(\pi_\epsilon(\bar{p}), \bar{p}),$$

$$\text{(A5)} \qquad \pi_\epsilon(\bar{p}) \;\in\; \arg\min_{p \in \mathcal{P}(\bar{p})} W_\epsilon^{\sigma_\epsilon}(p, \bar{p}).$$

Let $W_\epsilon(\pi_\epsilon, \bar{p}) \;:=\; W_\epsilon^{\sigma_\epsilon}(\pi_\epsilon, \bar{p})$ be the saddle point value of the ϵ-commitment solution at $\bar{p}$.

The ϵ-commitment saddle point value $W_\epsilon(\pi_\epsilon,\bar{p})$ is characterized as follows: For any $\delta \in (0,\epsilon)$,

$$
\begin{aligned}
W_\epsilon(\pi_\epsilon,\bar{p}) = \max_{m,\nu,\rho} m(\bar{p}) U_{\rho(\bar{p})}(\pi_\epsilon) + \left[1 - m(\bar{p})\right] \\
\times \Bigg\{ -\pi_\epsilon \int_{\bar{p}^\delta}^{\bar{p}} e^{-\int_0^{\tau(\bar{p},\bar{p}')}[\lambda+\nu(p^\tau)]d\tau} \left[-c + \lambda u_r^R + \nu(\bar{p}') u_{\rho(\bar{p}')}^R\right] \frac{1}{\eta(\bar{p}')} d\bar{p}' \\
- (1-\pi_\epsilon) \int_{\bar{p}^\delta}^{\bar{p}} e^{-\int_0^{\tau(\bar{p},\bar{p}')}\nu(p^\tau)d\tau} \left[-c + \nu(\bar{p}') u_{\rho(\bar{p}')}^L\right] \frac{1}{\eta(\bar{p}')} d\bar{p}' \\
+ \left[\pi_\epsilon e^{-\int_0^{\tau(\bar{p},\bar{p}^\delta)}[\lambda+\nu(p^\tau)]d\tau} + (1-\pi_\epsilon) e^{-\int_0^{\tau(\bar{p},\bar{p}^\delta)}\nu(p^\tau)d\tau}\right] W_{\epsilon-\delta}(\pi_\epsilon^\delta,\bar{p}^\delta) \Bigg\}.
\end{aligned}
$$

Following the dynamic programming principle, the RHS does not depend on any $\delta \in (0,\epsilon)$. Hence, the derivative of the RHS with respect to δ evaluated at $\delta = 0$ must equal zero. So, we obtain the standard HJB equations:[25]

$$\text{(A6)} \qquad 0 = \max_{m,\nu,\rho} G(m,\nu,\rho,\pi_\epsilon,\bar{p},W_\epsilon,dW_\epsilon),$$

$$\text{(A7)} \qquad \sigma_\epsilon(\bar{p}) = \left(m_\epsilon(\bar{p}),\nu_\epsilon(\bar{p}),\rho_\epsilon(\bar{p})\right) \in \arg\max_{m,\nu,\rho} G(m,\nu,\rho,\pi_\epsilon,\bar{p},W_\epsilon,dW_\epsilon).$$

Likewise, by the saddle point condition, $W_\epsilon(\pi_\epsilon,\bar{p})$ is also equivalently characterized as for any $\delta \in (0,\epsilon)$,

$$
\begin{aligned}
W_\epsilon(\pi_\epsilon,\bar{p}) = \min_{p} m_\epsilon(\bar{p}) U_{\rho_\epsilon(\bar{p})}(p) + \left[1 - m_\epsilon(\bar{p})\right] \\
\times \Bigg\{ -p \int_{\bar{p}^\delta}^{\bar{p}} e^{-\int_0^{\tau(\bar{p},\bar{p}')}[\lambda+\nu_\epsilon(p^\tau)]d\tau} \left[-c + \lambda u_r^R + \nu_\epsilon(\bar{p}') u_{\rho(\bar{p}')}^R\right] \frac{1}{\eta(\bar{p}')} d\bar{p}' \\
- (1-p) \int_{\bar{p}^\delta}^{\bar{p}} e^{-\int_0^{\tau(\bar{p},\bar{p}')}\nu_\epsilon(p^\tau)d\tau} \left[-c + \nu_\epsilon(\bar{p}') u_{\rho(\bar{p}')}^L\right] \frac{1}{\eta(\bar{p}')} d\bar{p}' \\
+ \left[p e^{-\int_0^{\tau(\bar{p},\bar{p}^\delta)}[\lambda+\nu_\epsilon(p^\tau)]d\tau} + (1-p) e^{-\int_0^{\tau(\bar{p},\bar{p}^\delta)}\nu_\epsilon(p^\tau)d\tau}\right] W_{\epsilon-\delta}^\sigma(p^\delta,\bar{p}^\delta) \Bigg\}.
\end{aligned}
$$

[25] Since

$$W_{\epsilon-\delta}(\pi_\epsilon^\delta,\bar{p}^\delta) = W_\epsilon(\pi_\epsilon,\bar{p}_-) + \frac{\partial W_\epsilon(\pi_\epsilon,\bar{p}_-)}{\partial p}\eta(\pi_\epsilon)\delta + \frac{\partial W_\epsilon(\pi_\epsilon,\bar{p}_-)}{\partial \bar{p}}\eta(\bar{p})\delta + o(\delta),$$

the derivative of $W_{\epsilon-\delta}(\pi_\epsilon^\delta,\bar{p}^\delta)$ with respect to δ is

$$\lim_{\delta\downarrow 0}\frac{W_{\epsilon-\delta}(\pi_\epsilon^\delta,\bar{p}^\delta) - W_\epsilon(\pi_\epsilon,\bar{p}_-)}{\delta} = \frac{\partial W_\epsilon(\pi_\epsilon,\bar{p}_-)}{\partial p}\eta(\pi_\epsilon) + \frac{\partial W_\epsilon(\pi_\epsilon,\bar{p}_-)}{\partial \bar{p}}\eta(\bar{p}).$$

Following the dynamic programming principle, the RHS does not depend on $\delta \in (0,\epsilon)$. Hence, the derivative of the RHS with respect to δ evaluated at $\delta = 0$ must equal zero. So, we obtain the standard HJB equations:

$$\pi_\epsilon \in \underset{p \in \mathcal{P}(\bar{p})}{\arg\min}\, G\big(m_\epsilon, \nu_\epsilon, \rho_\epsilon, p, \bar{p}, W_\epsilon, dW_\epsilon\big). \tag{A8}$$

The proof is in online Appendix B.4. We begin with a useful observation:

LEMMA 1: *Suppose a well-behaved pseudo equilibrium (σ^*, π^*) is dynamically credible. Then, for each $\bar{p} \in (0,1)$, for any sequence such that $\big(\sigma_\epsilon(\bar{p}), \pi_\epsilon(\bar{p})\big) \to \big(\sigma^*(\bar{p}), \pi^*(\bar{p})\big)$ as $\epsilon \to 0$, $W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}_-\big) \to V^{\sigma^*}\big(\pi^*(\bar{p}), \bar{p}_-\big)$, $\partial W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}_-\big)/\partial p \to V_p^{\sigma^*}\big(\pi^*(\bar{p}), \bar{p}_-\big)$, and $\partial W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}_-\big)/\partial \bar{p} \to V_{\bar{p}}^{\sigma^*}\big(\pi^*(\bar{p}), \bar{p}_-\big)$.*

The result now follows.

THEOREM 2: *If a well-behaved pseudo equilibrium is dynamically credible, then it is an intrapersonal equilibrium. Conversely, if there is a unique intrapersonal equilibrium, then it is dynamically credible.*

PROOF:

Fix any $\bar{p}$. By the dynamic credibility of (σ^*, π^*), for each $\bar{p} \in (0,1)$, there exists a sequence of ϵ such that an associated ϵ-commitment solution $\big(\sigma_\epsilon(\bar{p}), \pi_\epsilon(\bar{p})\big)$ converges to $\big(\sigma^*(\bar{p}), \pi^*(\bar{p})\big)$ as $\epsilon \to 0$. Given the continuity assumption, by Lemma 1, $W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}_-\big) \to V^{\sigma^*}\big(\pi^*(\bar{p}), \bar{p}_-\big)$, $\partial W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}_-\big)/\partial p \to V_p^{\sigma^*}\big(\pi^*(\bar{p}), \bar{p}_-\big)$, and $\partial W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}_-\big)/\partial \bar{p} \to V_{\bar{p}}^{\sigma^*}\big(\pi^*(\bar{p}), \bar{p}_-\big)$.

Next, observe that G is continuous. Hence, by Berge's maximum theorem, the maximized value is continuous and the set of maximizers must be upper hemicontinuous, which in this case should mean a closed graph. Hence, equations (11) and (12) must follow from equations (A6) and (A7) in the limit as $\epsilon \to 0$.

Applying the same argument, equation (13) follows from equation (A8) in the limit as $\epsilon \to 0$. Again, by Berge's maximum theorem, the minimized value is continuous and the set of minimizers must be upper hemicontinuous, which implies a closed graph. Hence, the conditions must hold in the limit as $\epsilon \to 0$.

Conversely, suppose (σ^*, π^*) is an intrapersonal equilibrium. Then, it satisfies equations (11), (12), and (13), for each $\bar{p}$. Hence, for each $\bar{p}$, $V^{\sigma^*}\big(\pi^*(\bar{p}), \bar{p}\big)$, $V_p^{\sigma^*}\big(\pi^*(\bar{p}), \bar{p}\big)$, and $V_p^{\sigma^*}\big(\pi^*(\bar{p}), \bar{p}_-\big)$ are well defined and continuous at $\big(\pi^*(\bar{p}), \bar{p}\big)$. Take a sequence (subsequence if necessary) of $\epsilon > 0$ such that $\Big(\sigma_\epsilon, \pi_\epsilon, W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}\big), \partial W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}\big)/\partial p, \partial W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}\big)/\partial \bar{p}\Big)$ all converge to some limit, say, $\Big(\sigma_0, \pi_0, W_0\big(\pi_0(\bar{p}), \bar{p}\big), \tilde{W}_{0,p}\big(\pi_0(\bar{p}), \bar{p}\big), \tilde{W}_{0,\bar{p}}\big(\pi_0(\bar{p}), \bar{p}\big)\Big)$. The reason that such a sequence can be found is as follows. First, $\Big(W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}\big), \partial W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}\big)/\partial p\Big)$ is bounded within a compact set $\Big[\min\big\{u_\ell^R, u_r^L\big\}, \max\big\{u_r^R, u_\ell^L\big\}\Big] \times \big[u_\ell^R - u_\ell^L, u_r^R - u_r^L\big]$. This together with equation (A7) means that $\nu_\epsilon(\bar{p})$ can be bounded within $[0, \bar{\nu}]$ for some large $\bar{\nu}$. The boundedness of $\nu_\epsilon(\bar{p})$ in turn means that $\partial W_\epsilon\big(\pi_\epsilon(\bar{p}), \bar{p}\big)/\partial \bar{p}$ can be bounded within a compact set. This proves that the converging sequence can be found for each $\bar{p}$. Next, since $\partial^2 W_\epsilon\big(\pi_\epsilon(\bar{p}),\bar{p}\big)/\partial p \partial \bar{p} = 0$ for all ϵ (which follows from the fact that $W_\epsilon(\cdot,\bar{p})$ is linear), we have $\tilde{W}_{0,p}\big(\pi_0(\bar{p}),\bar{p}\big) = \partial W_0\big(\pi_0(\bar{p}),\bar{p}\big)/\partial p$ and $\tilde{W}_{0,\bar{p}}\big(\pi_0(\bar{p}),\bar{p}\big) = \partial W_0\big(\pi_0(\bar{p}),\bar{p}\big)/\partial \bar{p}$. It is immediate that the limiting strategy profile (σ_0, π_0) is a pseudo equilibrium. Also using the same argument as above, we conclude that the limit strategy profile (σ_0, π_0), together with the value function W_0, satisfies equations (11), (12), and (13) for each $\bar{p}$. Therefore, (σ_0, π_0) is also an intrapersonal equilibrium. The uniqueness of the equilibrium then means that $(\sigma_0, \pi_0) = (\sigma^*, \pi^*)$, so for each $\bar{p}$, $\big(\sigma^*(\bar{p}), \pi_0^*(\bar{p})\big)$ is a limit of some ϵ-commitment solution $\big(\sigma_\epsilon(\bar{p}), \pi_\epsilon(\bar{p})\big)$ as $\epsilon \to 0$. Hence, (σ^*, π^*) is dynamically credible. ∎

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